

Biscayne Bay Aquatic Preserve

SEACAR Habitat Analyses

Last compiled on 10 June, 2026

Contents

Funding & Acknowledgements	2
Threshold Filtering	2
Value Qualifiers	3
Water Column	5
Seasonal Kendall-Tau Analysis	5
Water Quality - Discrete	5
Chlorophyll a, Corrected for Pheophytin - Discrete	6
Chlorophyll a, Uncorrected for Pheophytin - Discrete	7
Dissolved Oxygen - Discrete	10
Dissolved Oxygen Saturation - Discrete	12
pH - Discrete	13
Salinity - Discrete	16
Total Nitrogen - Discrete	18
Total Phosphorus - Discrete	21
Turbidity - Discrete	23
Water Temperature - Discrete	25
Water Quality - Continuous	28
pH - Continuous	30
Dissolved Oxygen Saturation - Continuous	31
Dissolved Oxygen - Continuous	32
Turbidity - Continuous	33
Water Temperature - Continuous	34
Salinity - Continuous	35
Specific Conductivity - Continuous	36
Submerged Aquatic Vegetation	37
Parameters	37
Species	37
Notes	37
SAV Water Column Analysis	42
Species list	44
References	45

Funding & Acknowledgements

The data used in this analysis is from the Export Standardized Tables in the SEACAR Data Discovery Interface (DDI). Documents and information available through the SEACAR DDI are owned by the data provider(s) and users are expected to provide appropriate credit following accepted citation formats. Users are encouraged to access data to maximize utilization of gained knowledge, reducing redundant research and facilitating partnerships and scientific innovation.

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Threshold Filtering

Threshold filters, following the guidance of Florida Department of Environmental Protection's (*FDEP*) Division of Environmental Assessment and Restoration (*DEAR*) are used to exclude specific results values from the SEACAR Analysis. Based on the threshold filters, Quality Assurance / Quality Control (*QAQC*) Flags are inserted into the *SEACAR_QAQCFlagCode* and *SEACAR_QAQC_Description* columns of the export data. The *Include* column indicates whether the *QAQC* Flag will also indicate that data are excluded from analysis. No data are excluded from the data export, but the analysis scripts can use the *Include* column to exclude data (1 to include, 0 to exclude).

Table 1: Continuous Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Dissolved Oxygen	mg/L	-0.000001	50
Dissolved Oxygen Saturation	%	-0.000001	500
Fluorescent Dissolved Organic Matter	QSE	-	-
Salinity	ppt	-0.000001	70
Specific Conductivity	mS/cm	-0.000001	200
Turbidity	NTU	-0.000001	4000
Water Temperature	Degrees C	-5.000000	45
pH	None	2.000000	14

Table 2: Discrete Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Ammonia, Un-ionized (NH3)	mg/L	-	-
Ammonium (NH4)	mg/L	-	-

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Corrected for Pheophytin	ug/L	-	-
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Colored Dissolved Organic Matter	PCU	-	-
Dissolved Oxygen	mg/L	-0.000001	25
Dissolved Oxygen Saturation	%	-0.000001	310
Fluorescent Dissolved Organic Matter	QSE	-	-
Light Extinction Coefficient	m ⁻¹	-	-
NO2+3, Filtered	mg/L	-	-
Nitrate (NO3)	mg/L	-	-
Nitrite (NO2)	mg/L	-	-
Nitrogen, inorganic	mg/L	-	-
Nitrogen, organic	mg/L	-	-
Phosphate, Filtered (PO4)	mg/L	-	-
Salinity	ppt	-0.000001	70
Secchi Depth	m	0.000001	50
Specific Conductivity	mS/cm	0.005000	100
Total Ammonia (N)	mg/L	-	-
Total Kjeldahl Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Phosphorus	mg/L	-	-
Total Suspended Solids	mg/L	-	-
Turbidity	NTU	-	-
Water Temperature	Degrees C	3.000000	40
pH	None	2.000000	13

Table 3: Quality Assurance Flags inserted based on threshold checks listed in Table 1 and 2

<i>SEACAR QAQC Description</i>	<i>Include</i>	<i>SEACAR QAQCFlagCode</i>
Exceeds maximum threshold	0	2Q
Below minimum threshold	0	4Q
Within threshold tolerance	1	6Q
No defined thresholds for this parameter	1	7Q

Value Qualifiers

Value qualifier codes included within the data are used to exclude certain results from the analysis. The data are retained in the data export files, but the analysis uses the *Include* column to filter the results.

STORET and WIN value qualifier codes

Value qualifier codes from *STORET* and *WIN* data are examined with the database and used to populate the *Include* column in data exports.

Table 4: Value Qualifier codes excluded from analysis

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>MDL</i>	<i>Description</i>
STORET-WIN	H	0	0	Value based on field kit determination; results may not be accurate
STORET-WIN	J	0	0	Estimated value
STORET-WIN	V	0	0	Analyte was detected at or above method detection limit
STORET-WIN	Y	0	0	Lab analysis from an improperly preserved sample; data may be inaccurate

Discrete Water Quality Value Qualifiers

The following value qualifiers are highlighted in the Discrete Water Quality section of this report. An exception is made for **Program 476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network** and data flagged with Value Qualifier **H** are included for this program only.

H - Value based on field kit determination; results may not be accurate. This code shall be used if a field screening test (e.g., field gas chromatograph data, immunoassay, or vendor-supplied field kit) was used to generate the value and the field kit or method has not been recognized by the Department as equivalent to laboratory methods.

I - The reported value is greater than or equal to the laboratory method detection limit but less than the laboratory practical quantitation limit.

Q - Sample held beyond the accepted holding time. This code shall be used if the value is derived from a sample that was prepared or analyzed after the approved holding time restrictions for sample preparation or analysis.

S - Secchi disk visible to bottom of waterbody. The value reported is the depth of the waterbody at the location of the Secchi disk measurement.

U - Indicates that the compound was analyzed for but not detected. This symbol shall be used to indicate that the specified component was not detected. The value associated with the qualifier shall be the laboratory method detection limit. Unless requested by the client, less than the method detection limit values shall not be reported

Systemwide Monitoring Program (SWMP) value qualifier codes

Value qualifier codes from the *SWMP* continuous program are examined with the database and used to populate the *Include* column in data exports. *SWMP* Qualifier Codes are indicated by *QualifierSource=SWMP*.

Table 5: SWMP Value Qualifier codes

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>Description</i>
SWMP	-1	1	Optional parameter not collected
SWMP	-2	0	Missing data
SWMP	-3	0	Data rejected due to QA/QC
SWMP	-4	0	Outside low sensor range
SWMP	-5	0	Outside high sensor range
SWMP	0	1	Passed initial QA/QC checks
SWMP	1	0	Suspect data
SWMP	2	1	Reserved for future use
SWMP	3	1	Calculated data: non-vented depth/level sensor correction for changes in barometric pressure
SWMP	4	1	Historical: Pre-auto QA/QC
SWMP	5	1	Corrected data

Water Column

The water column habitat extends from the water's surface to the bottom sediments, and it's where fish, dolphins, crabs and people swim! So much life makes its home in the water column that the health of marine and coastal ecosystems, as well as human economies, depend on the condition of this vulnerable habitat. Local patterns of rainfall, temperature, winds and currents can rapidly change the condition of the water column, while global influences such as [El Niño/La Niña](#), large-scale fluctuation in sea temperatures and climate change can have long-term effects. Inputs from the prosperity of our day-to-day lives including farming, mining and forestry, and emissions from power generation, automobiles and water treatment can also alter the health of the water column. Acting alone or together, each input can have complex and lasting effects on habitats and ecosystems.

SEACAR evaluates water column health with several essential parameters. These include nutrient surveys of nitrogen and phosphorus, and water quality assessments of salinity, dissolved oxygen, pH, and water temperature. Water clarity is evaluated with Secchi depth, turbidity, levels of chlorophyll a, total suspended solids, and colored dissolved organic matter. Additionally, the richness of nekton is indicated by the abundance of free-swimming fishes and macroinvertebrates like crabs and shrimps.

Seasonal Kendall-Tau Analysis

Indicators must have a minimum of five to ten years, depending on the habitat, of data within the geographic range of the analysis to be included in the analysis. Ten years of data are required for discrete parameters, and five years of data are required for continuous parameters. If there are insufficient years of data, the number of years of data available will be noted and labeled as “insufficient data to conduct analysis”. Further, for the preferred Seasonal Kendall-Tau test, there must be data from at least two months in common across at least two consecutive years within the RCP managed area being analyzed. Values that pass both of these tests will be included in the analysis and be labeled as *Use_In_Analysis* = **TRUE**. Any that fail either test will be excluded from the analyses and labeled as *Use_In_Analysis* = **FALSE**. The points for all Water Column plots displayed in this section are monthly averages. Trend significance will be denoted as “Significant Trend” (when $p < 0.05$), or “Non-significant Trend” (when $p \geq 0.05$). Any parameters with insufficient data to perform Seasonal Kendall-Tau test will have their monthly averages plotted without a corresponding trend line.

Water Quality - Discrete

The following files were used in the discrete analysis:

- *Combined_WQ_WC_NUT_Chlorophyll_a_corrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Chlorophyll_a_uncorrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Colored_dissolved_organic_matter_CDOM-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen_Saturation-2026-May-18.txt*
- *Combined_WQ_WC_NUT_pH-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Salinity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Secchi_Depth-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Nitrogen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Phosphorus-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Suspended_Solids_TSS-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Turbidity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Water_Temperature-2026-May-18.txt*

Chlorophyll a, Corrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

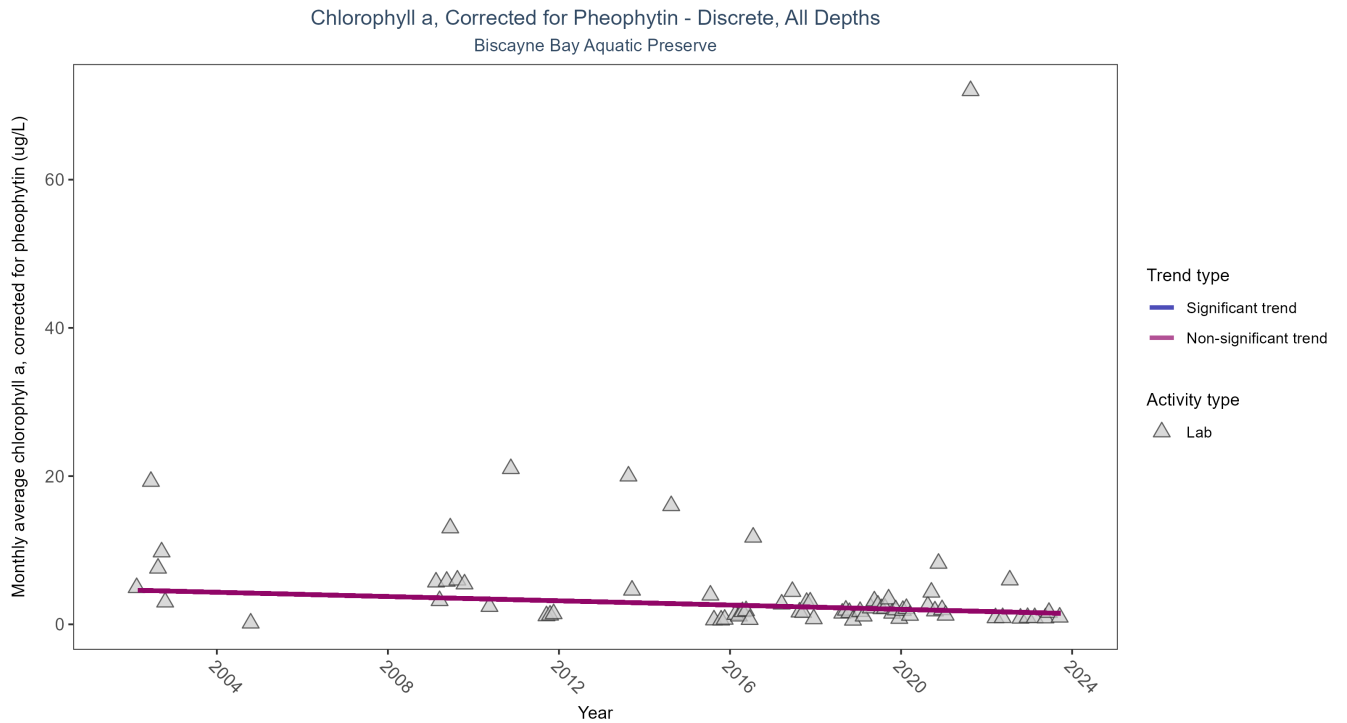


Figure 1: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 6: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	492	16	2002 - 2023	1.4	-0.2016	4.6156	-0.1439	0.0505

Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2002 and 2023.

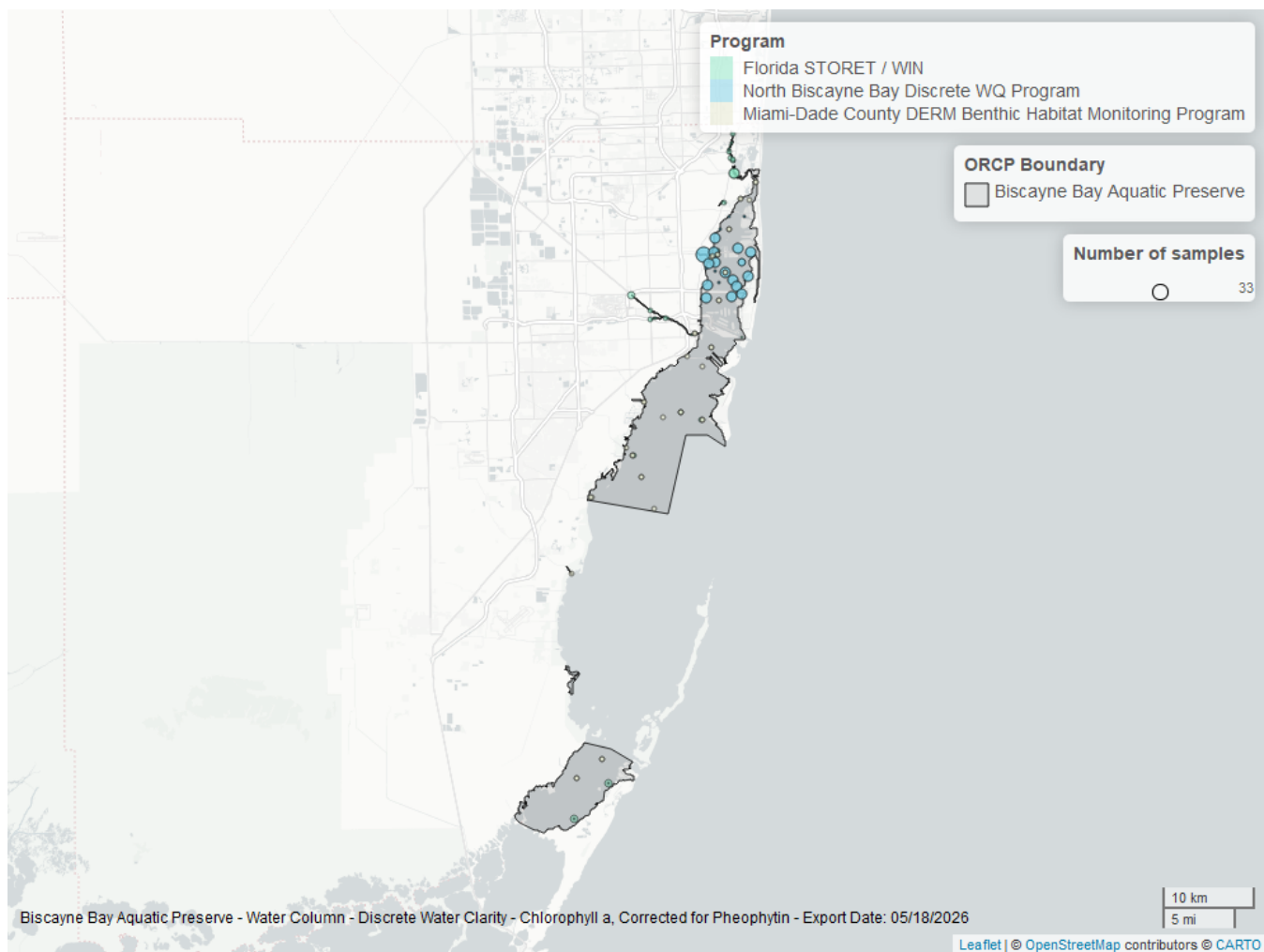


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 7: Programs contributing data for Chlorophyll a, Corrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
5026	287	2019	2021
5002	124	2002	2023
4018	82	2004	2011

Program names:

[5026](#) - North Biscayne Bay Discrete Water Quality Program¹

5002 - Florida STORET / WIN²

4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³

Chlorophyll a, Uncorrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

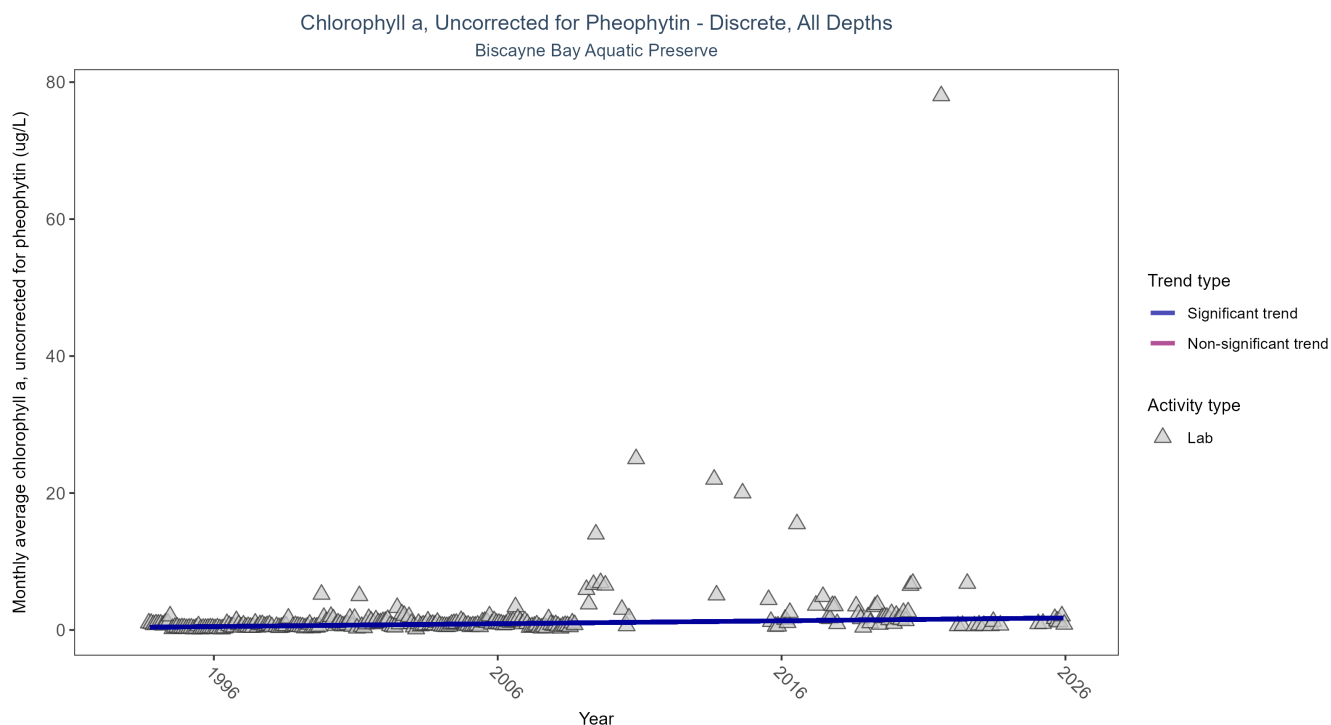


Figure 3: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 8: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	2517	30	1993 - 2025	0.7023	0.3739	0.374	0.0421	0

Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.04 $\mu\text{g/L}$ per year, indicating a decrease in water clarity.

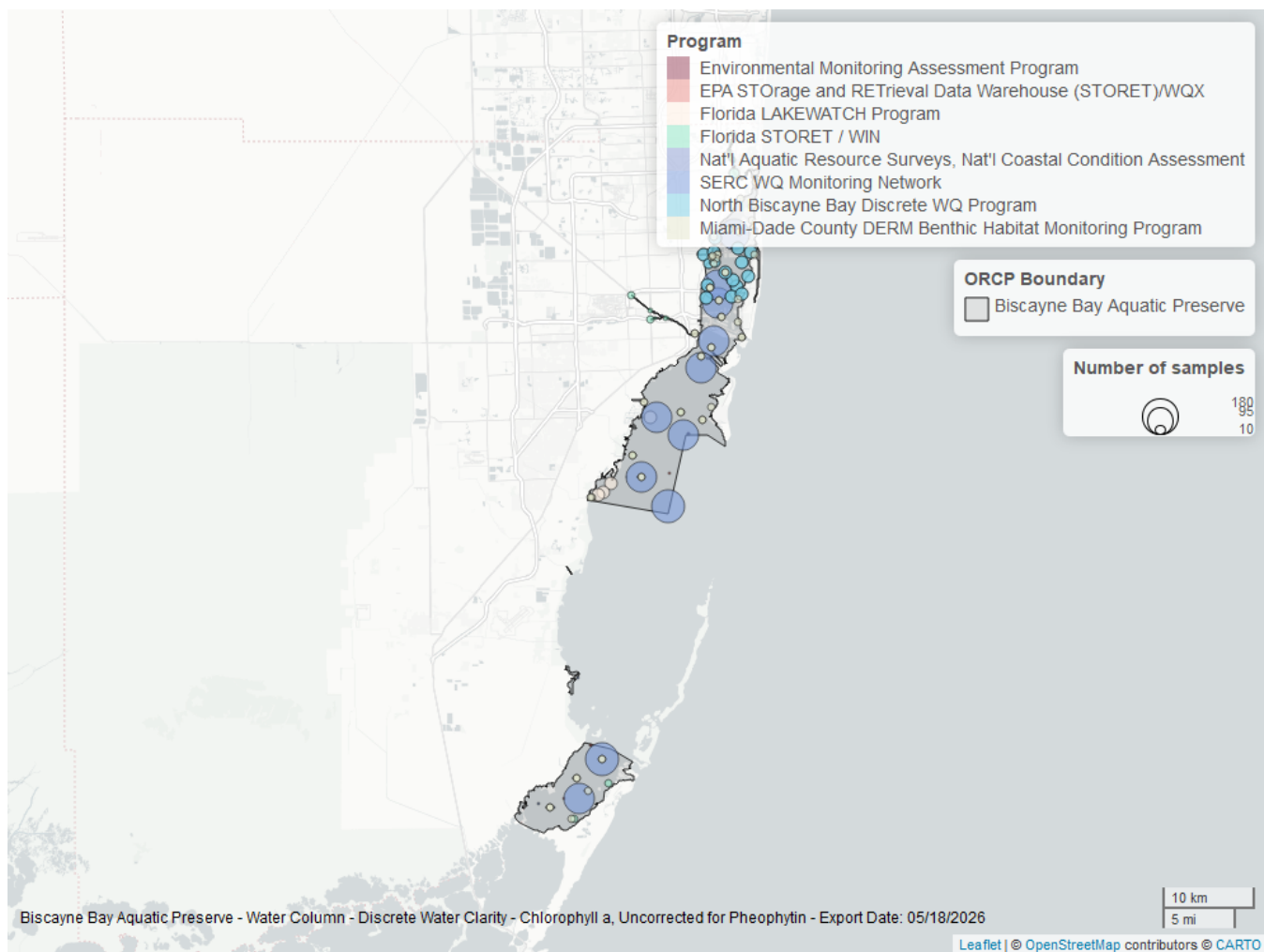


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 9: Programs contributing data for Chlorophyll a, Uncorrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
509	1636	1993	2008
5026	410	2019	2020
4018	247	2025	2025
5002	117	2001	2023
514	92	2000	2005
103	9	2002	2015
118	6	2006	2010
115	1	2004	2004

Program names:

509 - SERC Water Quality Monitoring Network⁴

5026 - North Biscayne Bay Discrete Water Quality Program¹

4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³

5002 - Florida STORET / WIN²

514 - Florida LAKEWATCH Program⁵

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁶
 118 - National Aquatic Resource Surveys, National Coastal Condition Assessment⁷
 115 - Environmental Monitoring Assessment Program⁸

Dissolved Oxygen - Discrete

Seasonal Kendall-Tau Trend Analysis

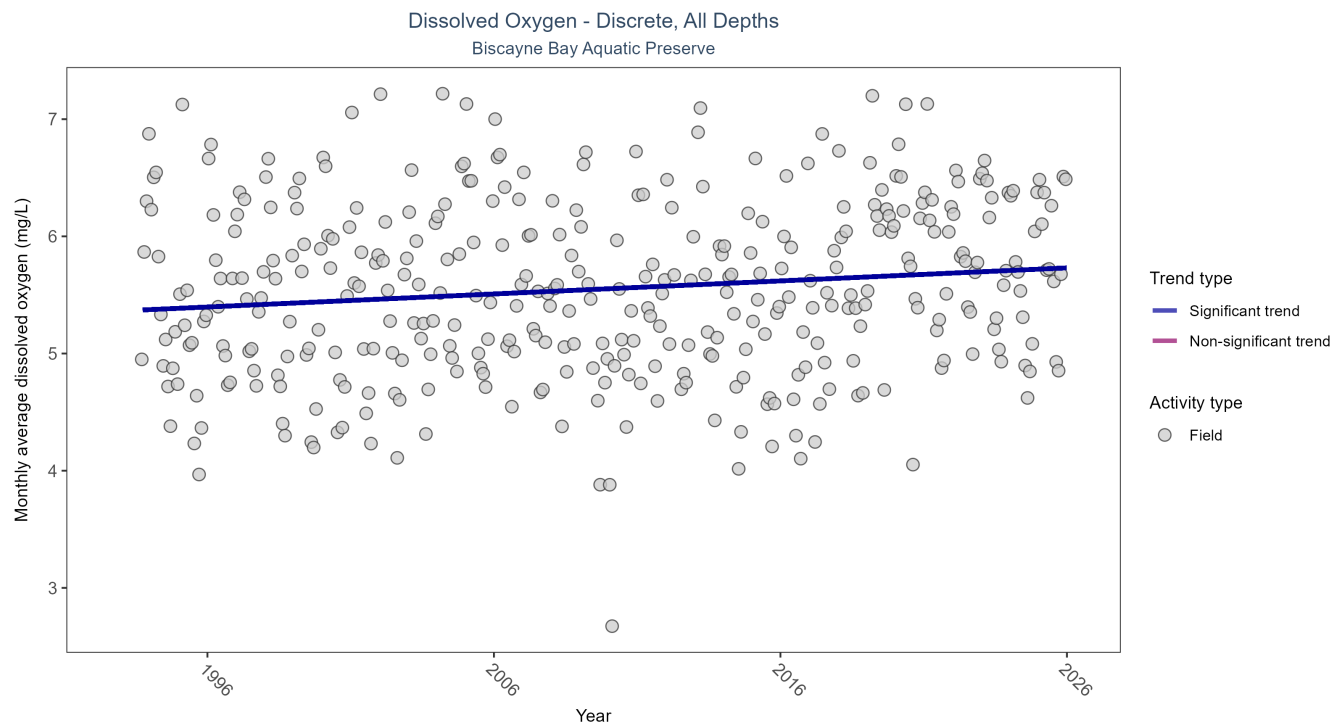


Figure 5: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 10: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	45505	33	1993 - 2025	5.81	0.179	5.3636	0.0111	0

Monthly average dissolved oxygen increased by 0.01 mg/L per year.

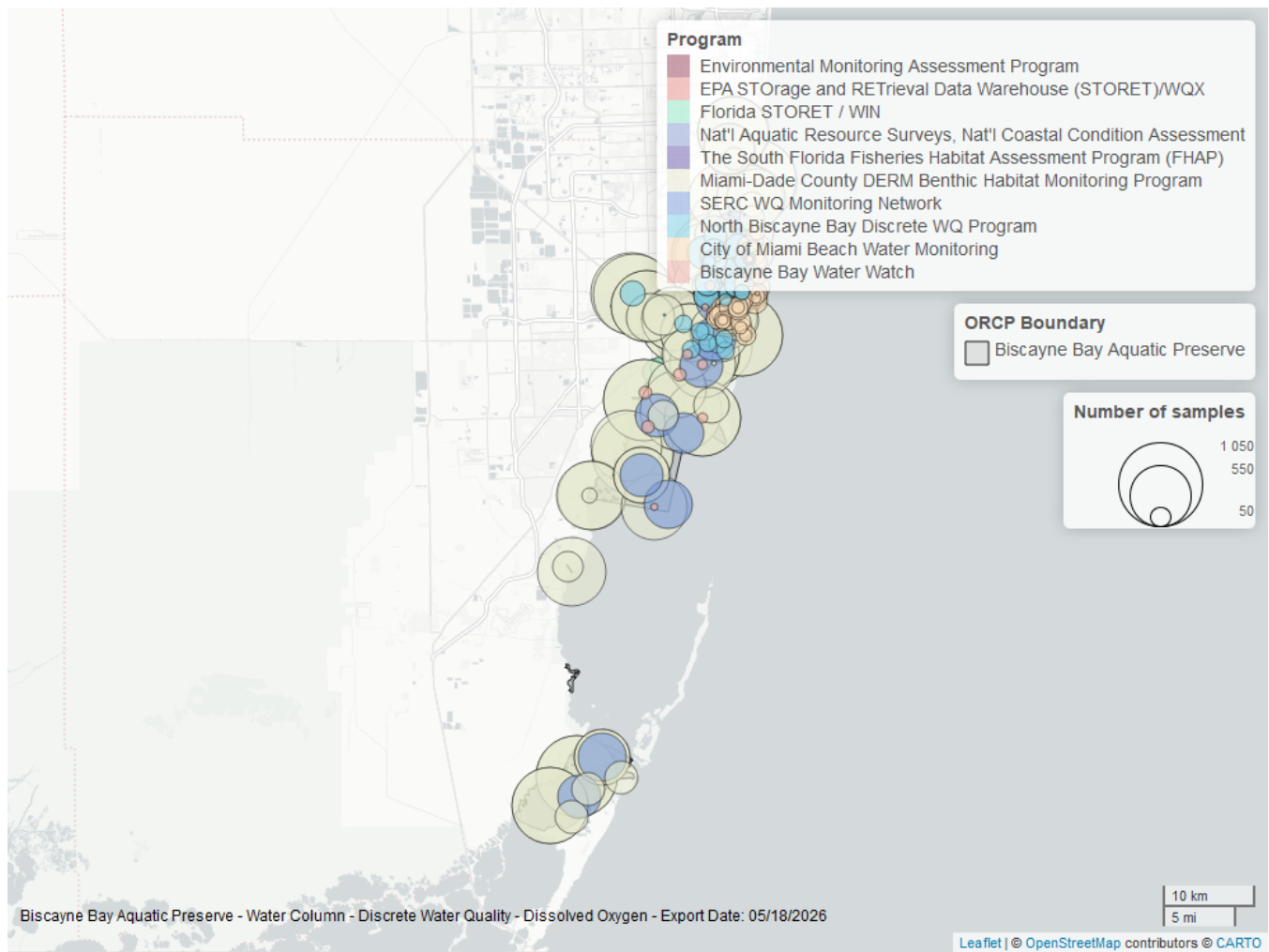


Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 11: Programs contributing data for Dissolved Oxygen

ProgramID	N_Data	YearMin	YearMax
4018	37475	1994	2025
509	3316	1993	2008
4058	2958	2016	2025
5026	1906	2019	2025
5002	1128	2001	2023
4049	192	2006	2008
4057	166	2015	2019
118	28	2006	2020
103	15	2015	2015
115	3	2004	2004

Program names:

4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³

509 - SERC Water Quality Monitoring Network⁴

4058 - City of Miami Beach Water Monitoring⁹

5026 - North Biscayne Bay Discrete Water Quality Program¹
5002 - Florida STORET / WIN²
4049 - The South Florida Fisheries Habitat Assessment Program (FHAP)¹⁰
4057 - Biscayne Bay Water Watch¹¹
118 - National Aquatic Resource Surveys, National Coastal Condition Assessment⁷
103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁶
115 - Environmental Monitoring Assessment Program⁸

Dissolved Oxygen Saturation - Discrete

Seasonal Kendall-Tau Trend Analysis

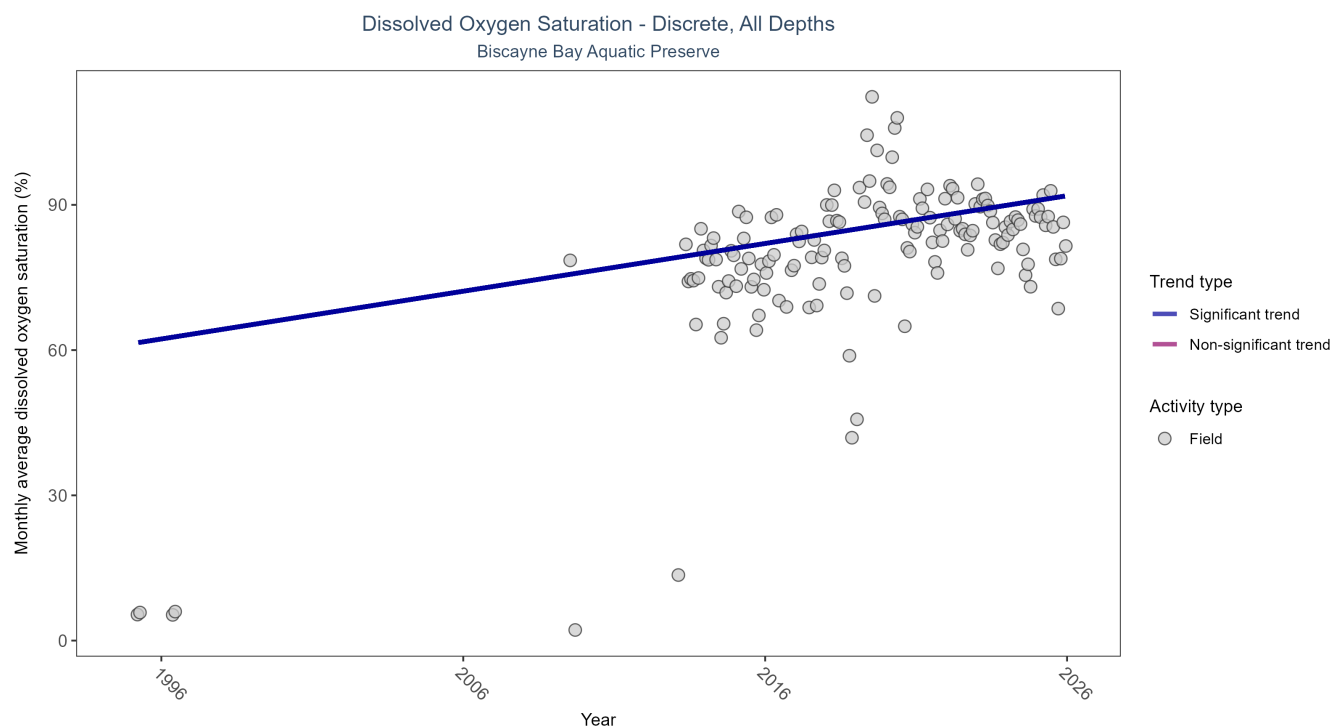


Figure 7: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen Saturation

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	15093	16	1995 - 2025	88.7	0.3847	61.3128	0.9861	0

Monthly average dissolved oxygen saturation increased by 0.99% per year.

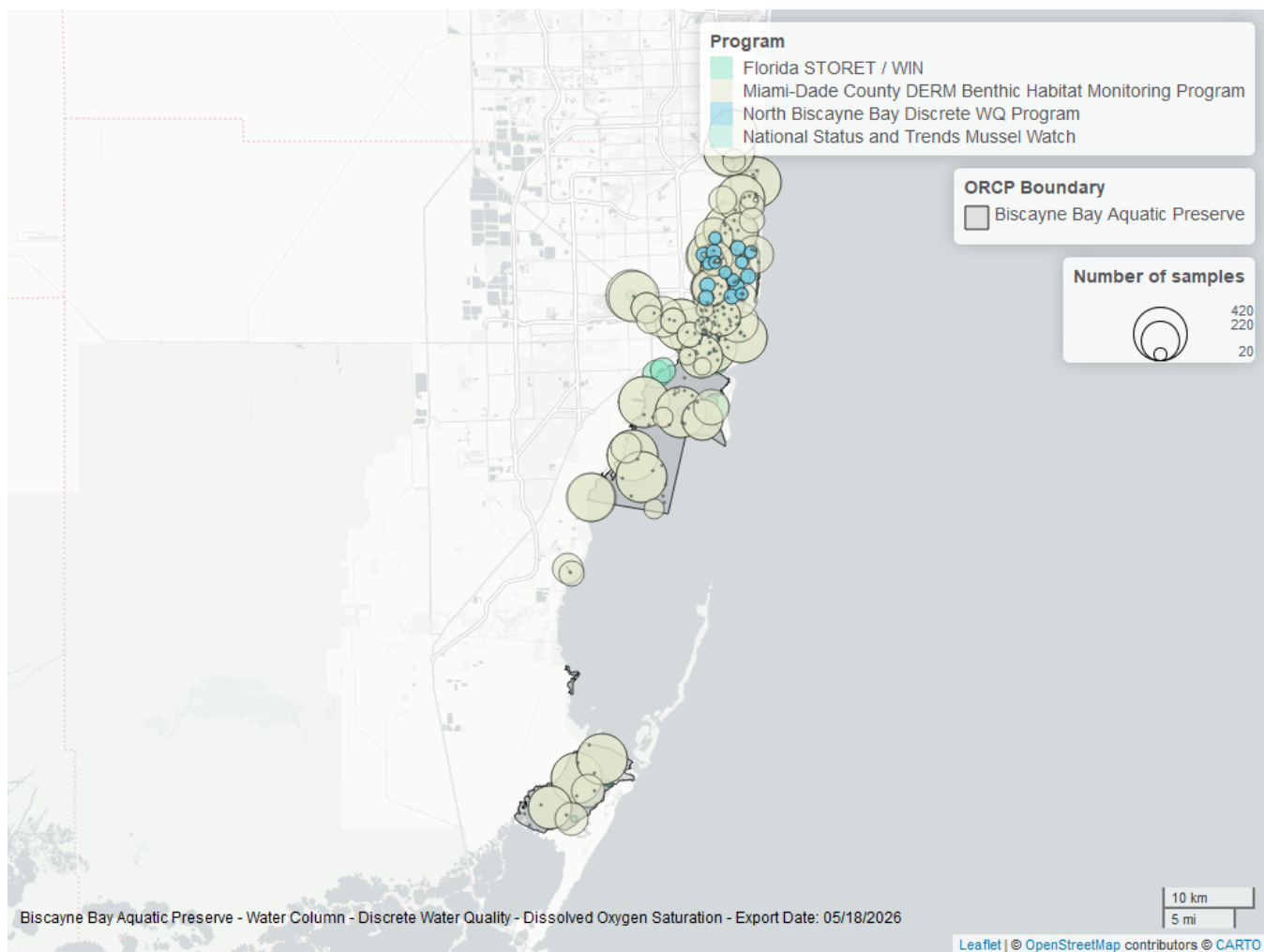


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 13: Programs contributing data for Dissolved Oxygen Saturation

ProgramID	N_Data	YearMin	YearMax
4018	13641	2009	2025
5002	889	2009	2023
5026	474	2019	2020
102	263	1995	1996

Program names:

4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³

5002 - Florida STORET / WIN²

5026 - North Biscayne Bay Discrete Water Quality Program¹

102 - National Status and Trends Mussel Watch¹²

pH - Discrete

Seasonal Kendall-Tau Trend Analysis

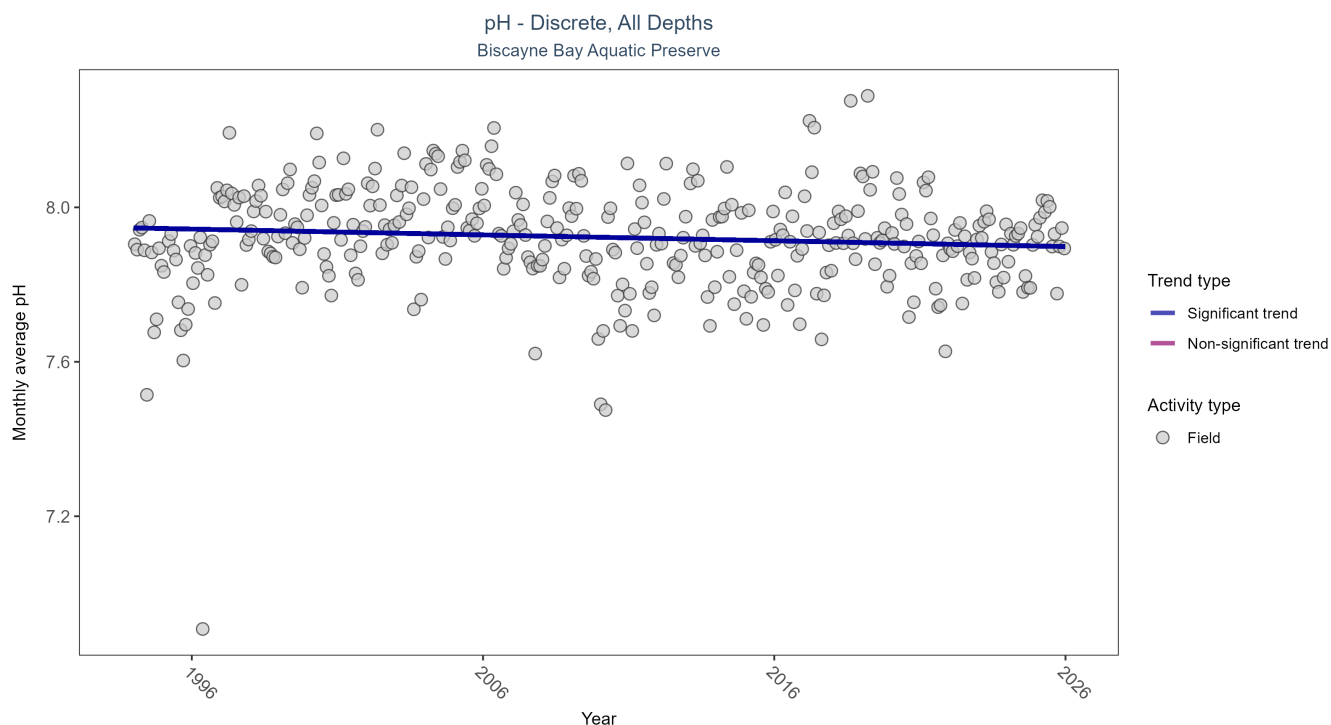


Figure 9: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 14: Seasonal Kendall-Tau Trend Analysis for pH

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	43873	32	1994 - 2025	7.98	-0.1044	7.9468	-0.0015	0.0038

Monthly average pH decreased by less than 0.01 pH units per year.

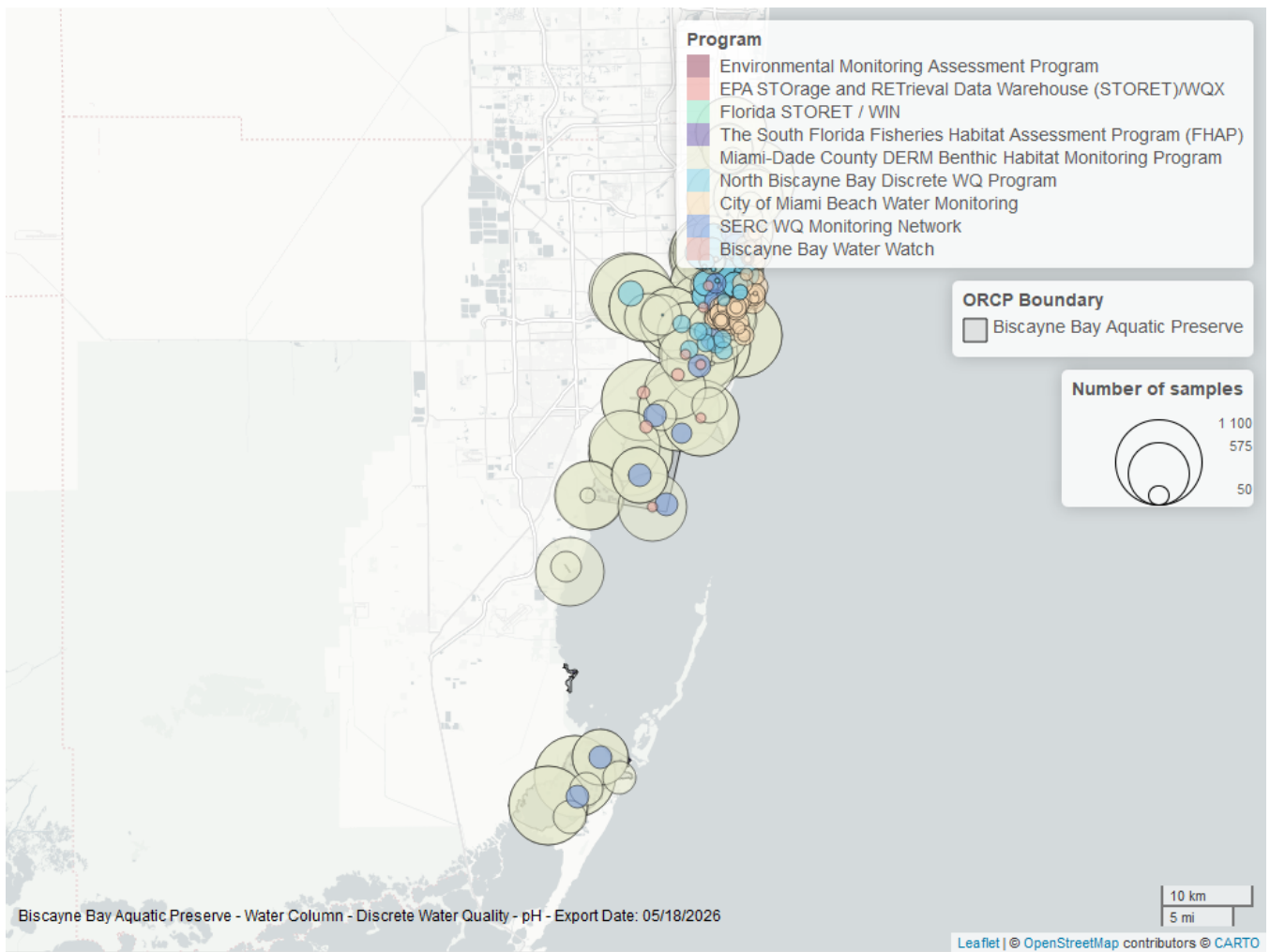


Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 15: Programs contributing data for pH

ProgramID	N_Data	YearMin	YearMax
4018	38513	1994	2025
4058	3002	2016	2025
5026	1975	2019	2025
509	822	2002	2008
5002	380	2001	2023
4049	247	2005	2008
4057	169	2015	2019
103	13	2015	2015
115	3	2004	2004

Program names:

4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³

4058 - City of Miami Beach Water Monitoring⁹

5026 - North Biscayne Bay Discrete Water Quality Program¹

509 - SERC Water Quality Monitoring Network⁴

5002 - Florida STORET / WIN²
4049 - The South Florida Fisheries Habitat Assessment Program (FHAP)¹⁰
4057 - Biscayne Bay Water Watch¹¹
103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁶
115 - Environmental Monitoring Assessment Program⁸

Salinity - Discrete

Seasonal Kendall-Tau Trend Analysis

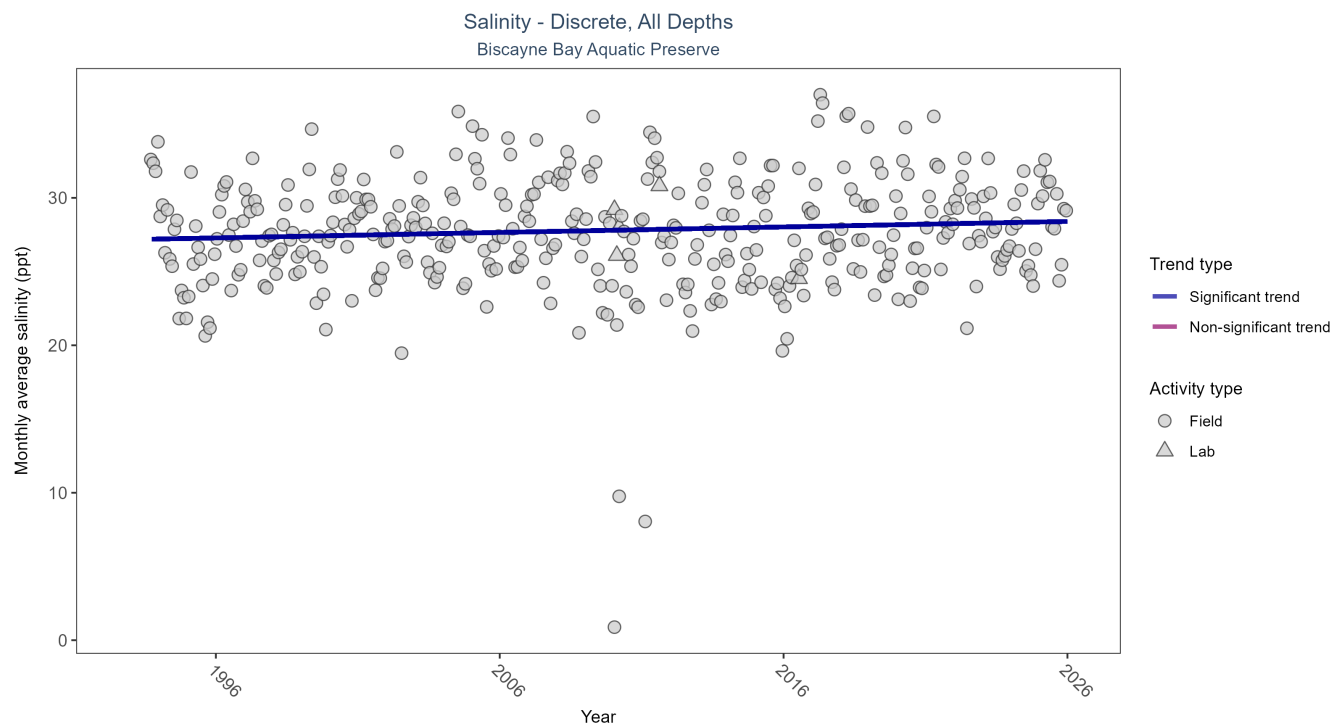


Figure 11: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 16: Seasonal Kendall-Tau Trend Analysis for Salinity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Increasing trend	50500	33	1993 - 2025	31.4	0.0877	27.1694	0.0372	0.0144

Monthly average salinity increased by 0.04 ppt per year.

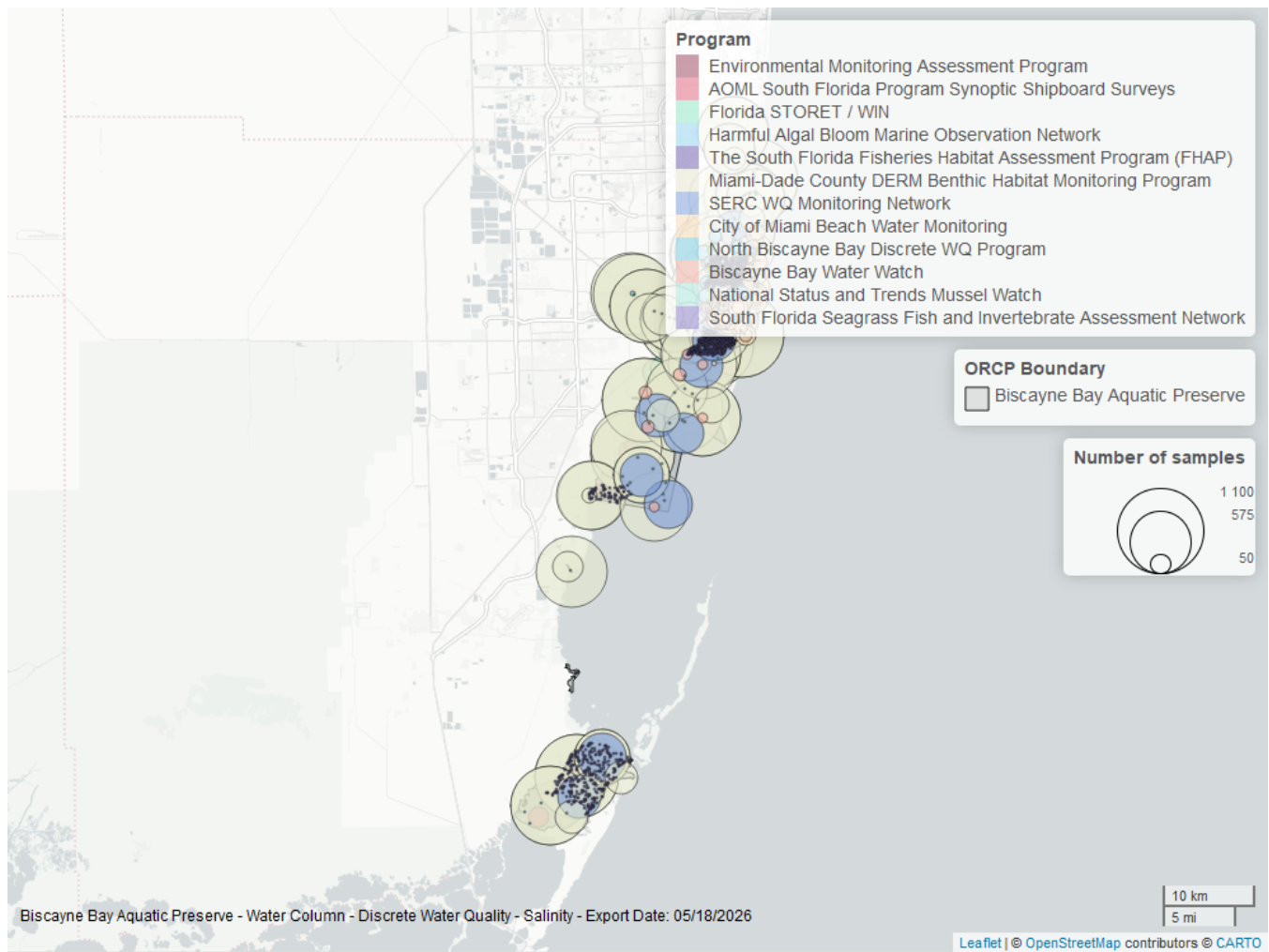


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 17: Programs contributing data for Salinity

ProgramID	N_Data	YearMin	YearMax
4018	39125	1994	2025
509	3316	1993	2008
4058	2996	2016	2025
965	2514	2005	2011
5002	1087	2009	2023
5026	465	2019	2025
3	392	2002	2012
4049	271	2005	2008
102	267	1995	1996
4057	171	2015	2019
95	7	2013	2013
115	3	2004	2004

Program names:

4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³

509 - SERC Water Quality Monitoring Network⁴

4058 - City of Miami Beach Water Monitoring⁹

965 - South Florida Seagrass Fish and Invertebrate Assessment Network¹³

5002 - Florida STORET / WIN²

5026 - North Biscayne Bay Discrete Water Quality Program¹

3 - Atlantic Oceanographic and Meteorological Laboratory (AOML) South Florida Program Synoptic Shipboard Surveys¹⁴

4049 - The South Florida Fisheries Habitat Assessment Program (FHAP)¹⁰

102 - National Status and Trends Mussel Watch¹²

4057 - Biscayne Bay Water Watch¹¹

95 - Harmful Algal Bloom Marine Observation Network¹⁵

115 - Environmental Monitoring Assessment Program⁸

Total Nitrogen - Discrete

Total Nitrogen Calculation:

The logic for calculated Total Nitrogen was provided by Kevin O'Donnell and colleagues at FDEP (with the help of Jay Silvanima, Watershed Monitoring Section). The following logic is used, in this order, based on the availability of specific nitrogen components.

- 1) $TN = TKN + NO_3O_2$;
- 2) $TN = TKN + NO_3 + NO_2$;
- 3) $TN = ORGN + NH_4 + NO_3O_2$;
- 4) $TN = ORGN + NH_4 + NO_2 + NO_3$;
- 5) $TN = TKN + NO_3$;
- 6) $TN = ORGN + NH_4 + NO_3$;

Additional Information:

- Rules for use of sample fraction:
 - Florida Department of Environmental Protection (FDEP) report that if both “Total” and “Dissolved” components are reported, only “Total” is used. If the total is not reported, then the dissolved components are used as a best available replacement.
 - Total nitrogen calculations are done using nitrogen components with the same sample fraction, nitrogen components with mixed total/dissolved sample fractions are not used. In other words, total nitrogen can be calculated when TKN and NO_3O_2 are both total sample fractions, or when both are dissolved sample fractions. *Future calculations of total nitrogen values may be based on components with mixed sample fractions.*
- Values inserted into data:
 - ParameterName = “Total Nitrogen”
 - SEACAR_QAQCFlagCode = “1Q”
 - SEACAR_QAQC_Description = “SEACAR Calculated”

Seasonal Kendall-Tau Trend Analysis

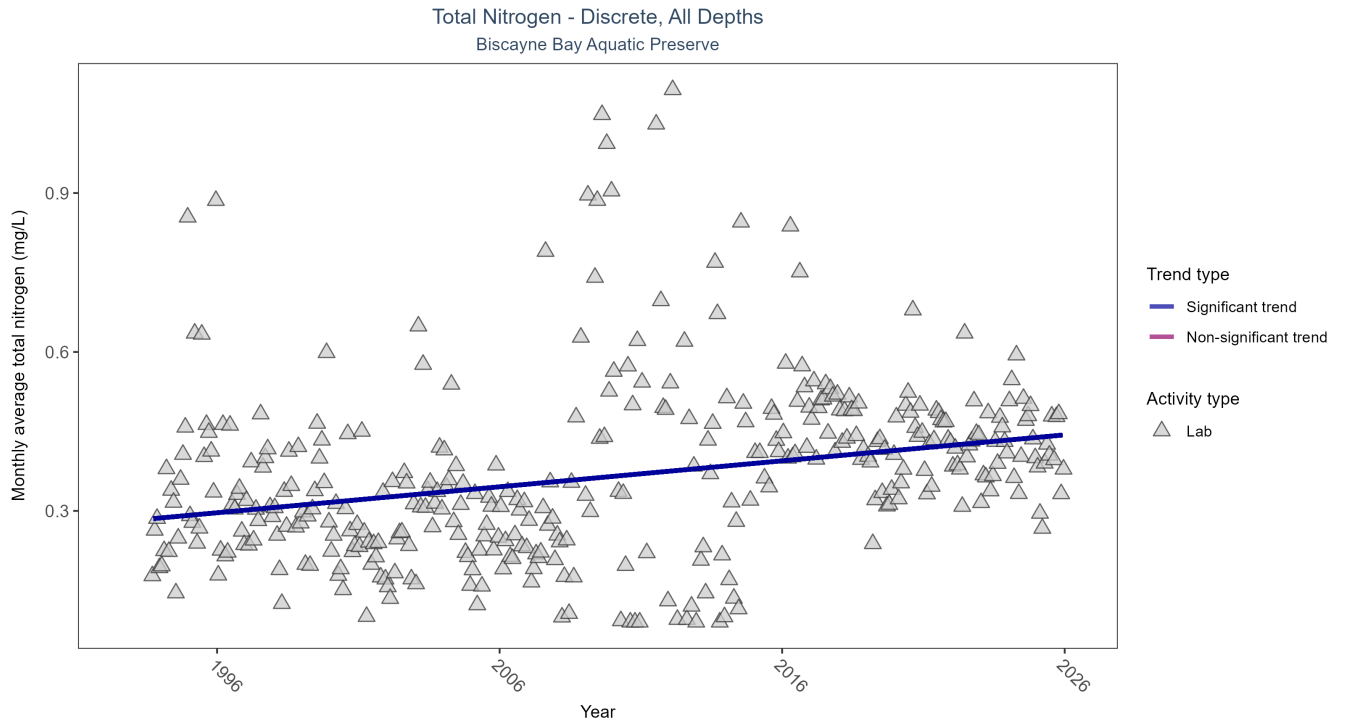


Figure 13: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 18: Seasonal Kendall-Tau Trend Analysis for Total Nitrogen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	9363	33	1993 - 2025	0.335	0.2387	0.2818	0.0049	0

Monthly average total nitrogen increased by less than 0.01 mg/L per year.

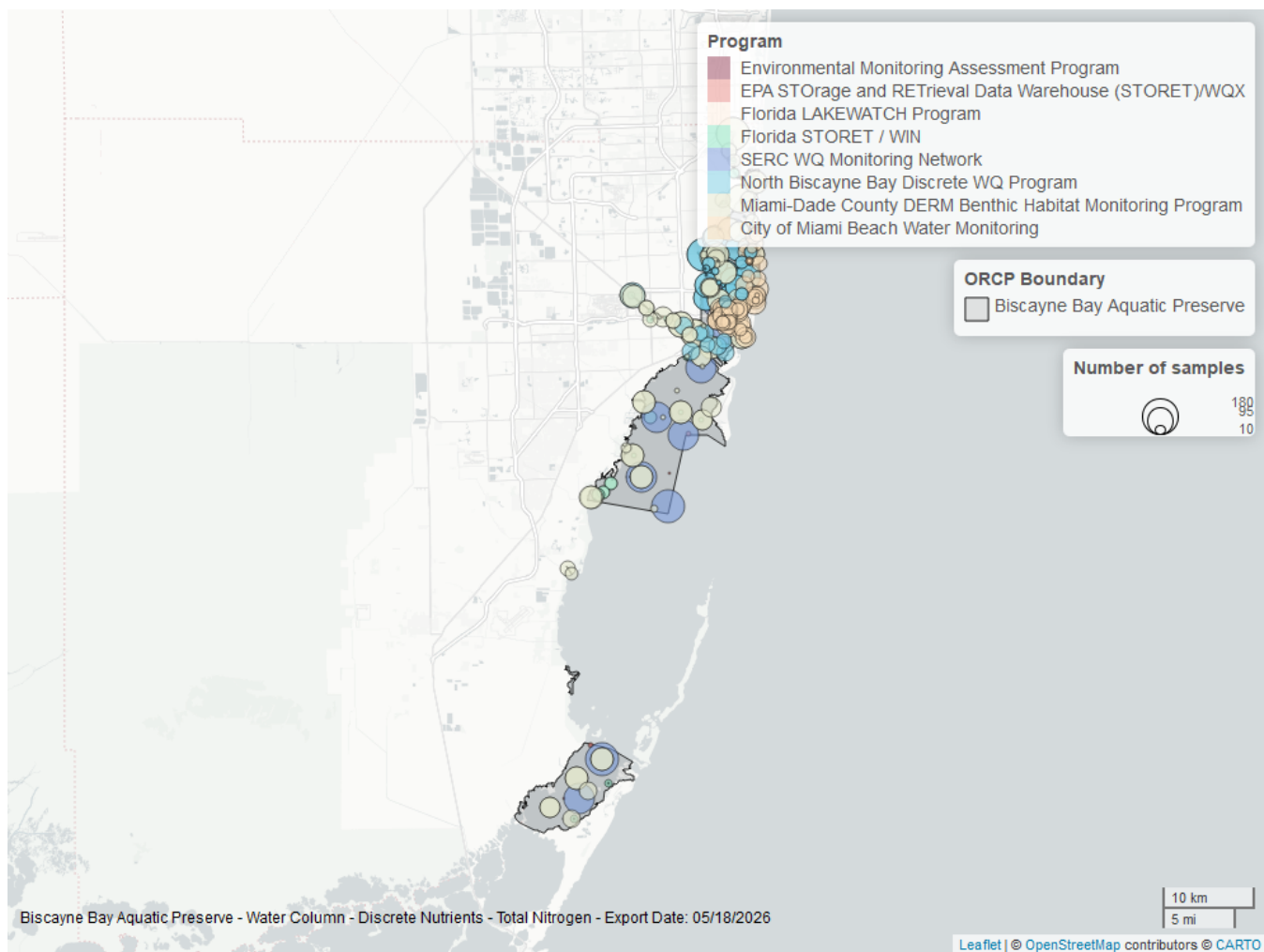


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 19: Programs contributing data for Total Nitrogen

ProgramID	N_Data	YearMin	YearMax
4018	3222	1994	2025
4058	2359	2016	2025
5026	1831	2019	2025
509	1654	1993	2008
5002	236	2000	2023
514	104	2000	2005
103	19	2002	2015
115	1	2004	2004

Program names:

4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³

4058 - City of Miami Beach Water Monitoring⁹

5026 - North Biscayne Bay Discrete Water Quality Program¹

509 - SERC Water Quality Monitoring Network⁴

5002 - Florida STORET / WIN²

514 - Florida LAKEWATCH Program⁵
 103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁶
 115 - Environmental Monitoring Assessment Program⁸

Total Phosphorus - Discrete

Seasonal Kendall-Tau Trend Analysis

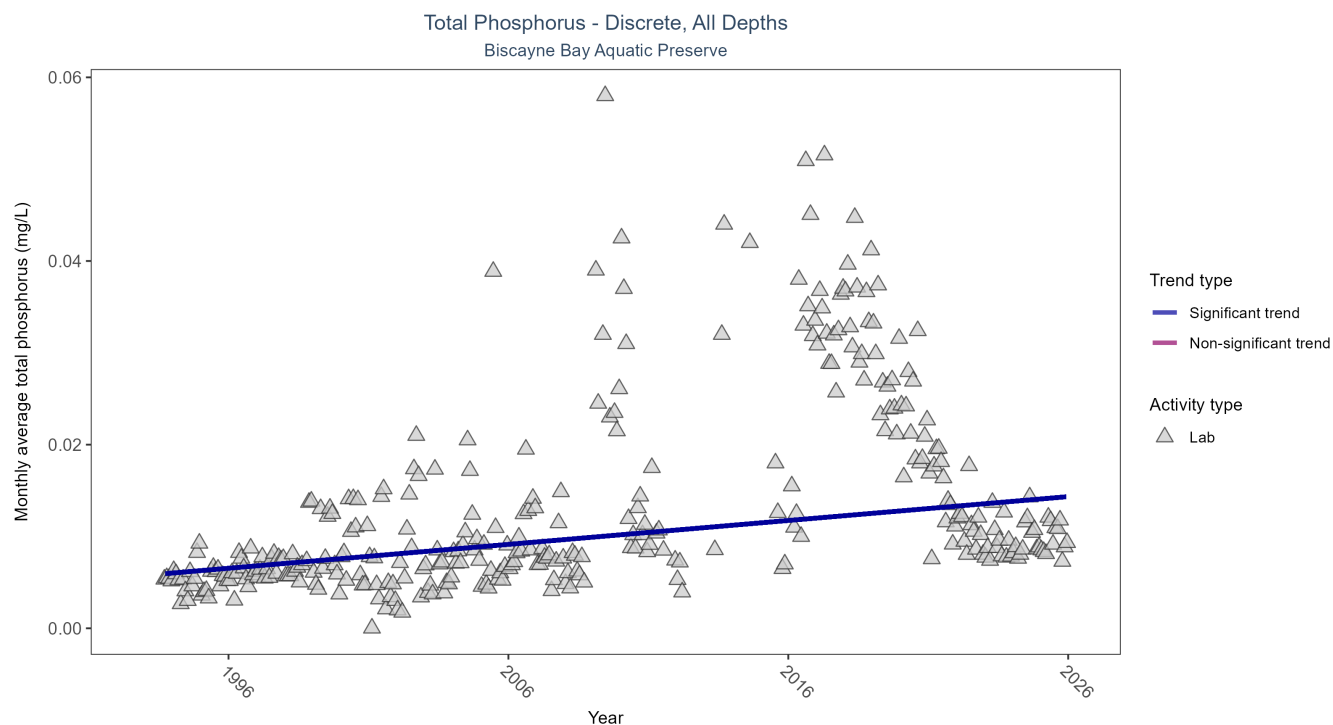


Figure 15: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 20: Seasonal Kendall-Tau Trend Analysis for Total Phosphorus

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	9858	33	1993 - 2025	0.009	0.3798	0.0058	0.0003	0

Monthly average total phosphorus increased by less than 0.01 mg/L per year.

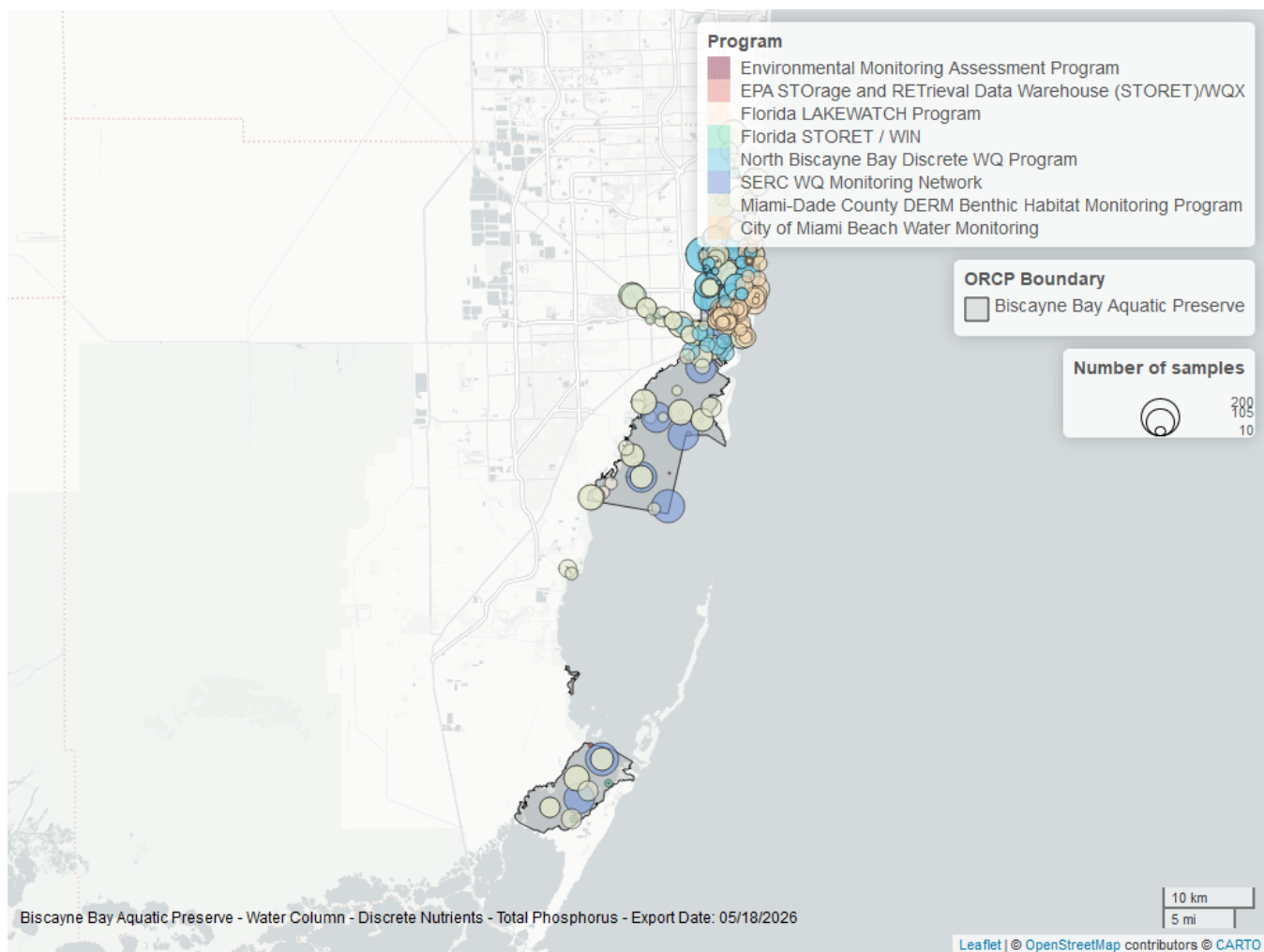


Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 21: Programs contributing data for Total Phosphorus

ProgramID	N_Data	YearMin	YearMax
4018	3655	2004	2025
4058	2523	2016	2025
5026	1896	2019	2025
509	1655	1993	2008
5002	171	2001	2023
514	103	2000	2005
103	15	2002	2015
115	1	2004	2004

Program names:

4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³

4058 - City of Miami Beach Water Monitoring⁹

5026 - North Biscayne Bay Discrete Water Quality Program¹

509 - SERC Water Quality Monitoring Network⁴

5002 - Florida STORET / WIN²

514 - Florida LAKEWATCH Program⁵
 103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁶
 115 - Environmental Monitoring Assessment Program⁸

Turbidity - Discrete

Seasonal Kendall-Tau Trend Analysis

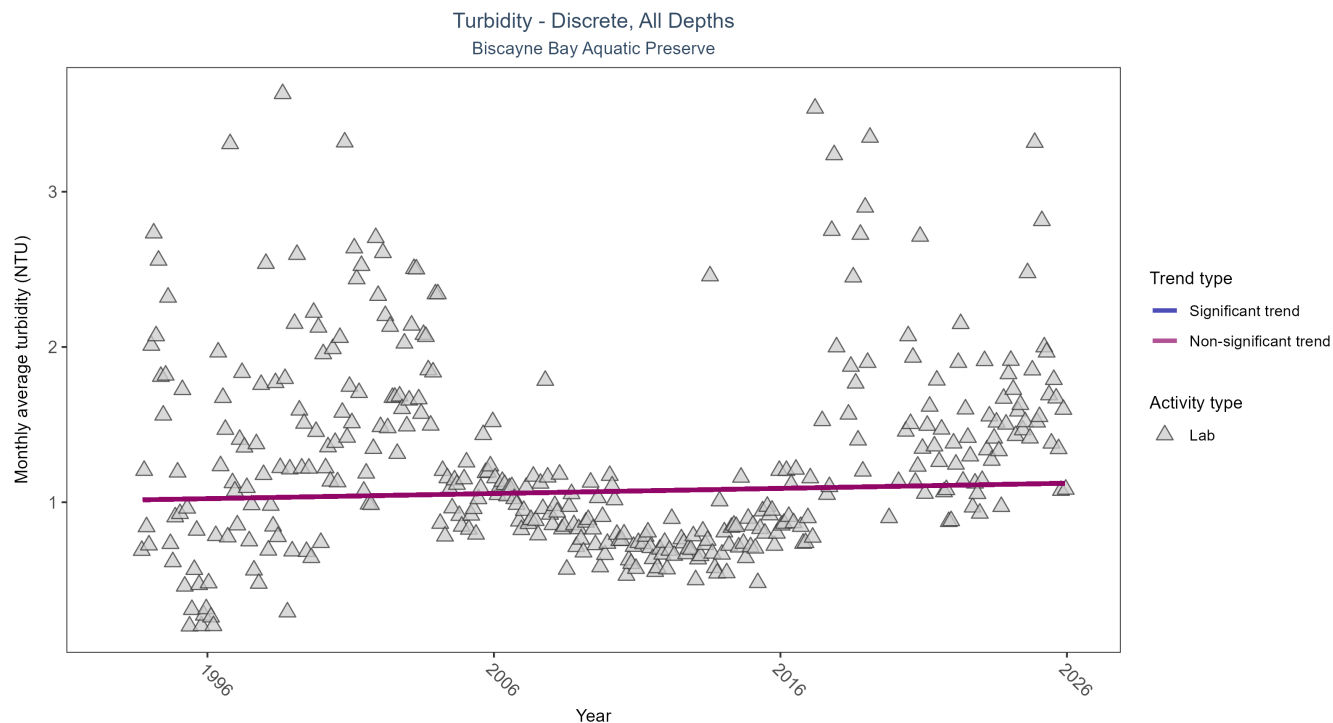


Figure 17: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 22: Seasonal Kendall-Tau Trend Analysis for Turbidity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	13958	33	1993 - 2025	0.9	0.04	1.0132	0.0033	0.2562

Turbidity showed no detectable trend between 1993 and 2025.

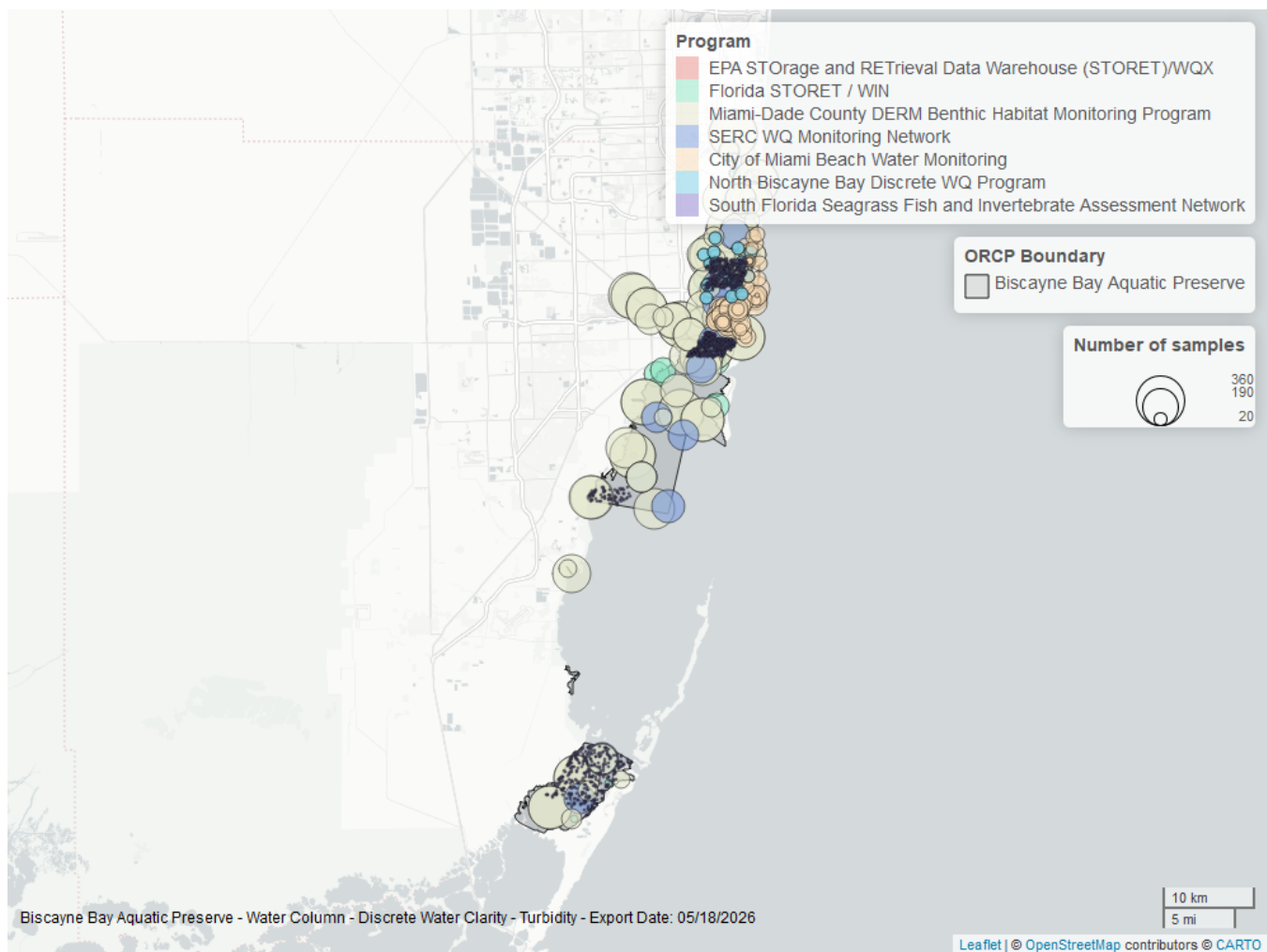


Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 23: Programs contributing data for Turbidity

ProgramID	N_Data	YearMin	YearMax
4018	12307	1994	2025
4058	3002	2016	2025
509	1658	1993	2008
965	1255	2005	2011
5002	876	2002	2023
5026	410	2019	2020
103	4	2006	2006

Program names:

4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³

4058 - City of Miami Beach Water Monitoring⁹

509 - SERC Water Quality Monitoring Network⁴

965 - South Florida Seagrass Fish and Invertebrate Assessment Network¹³

5002 - Florida STORET / WIN²

Water Temperature - Discrete

Seasonal Kendall-Tau Trend Analysis

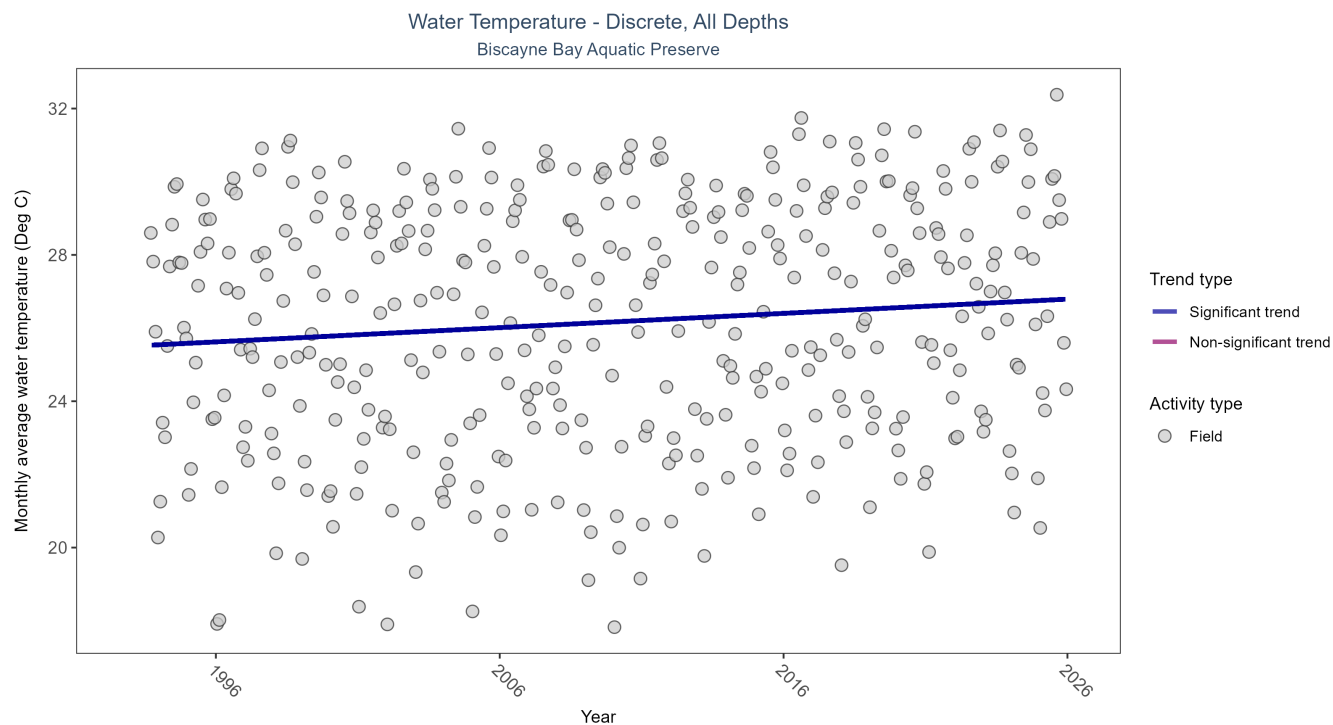


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 24: Seasonal Kendall-Tau Trend Analysis for Water Temperature

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	50728	33	1993 - 2025	26.63	0.2277	25.5048	0.039	0

Monthly average water temperature increased by 0.04°C per year.

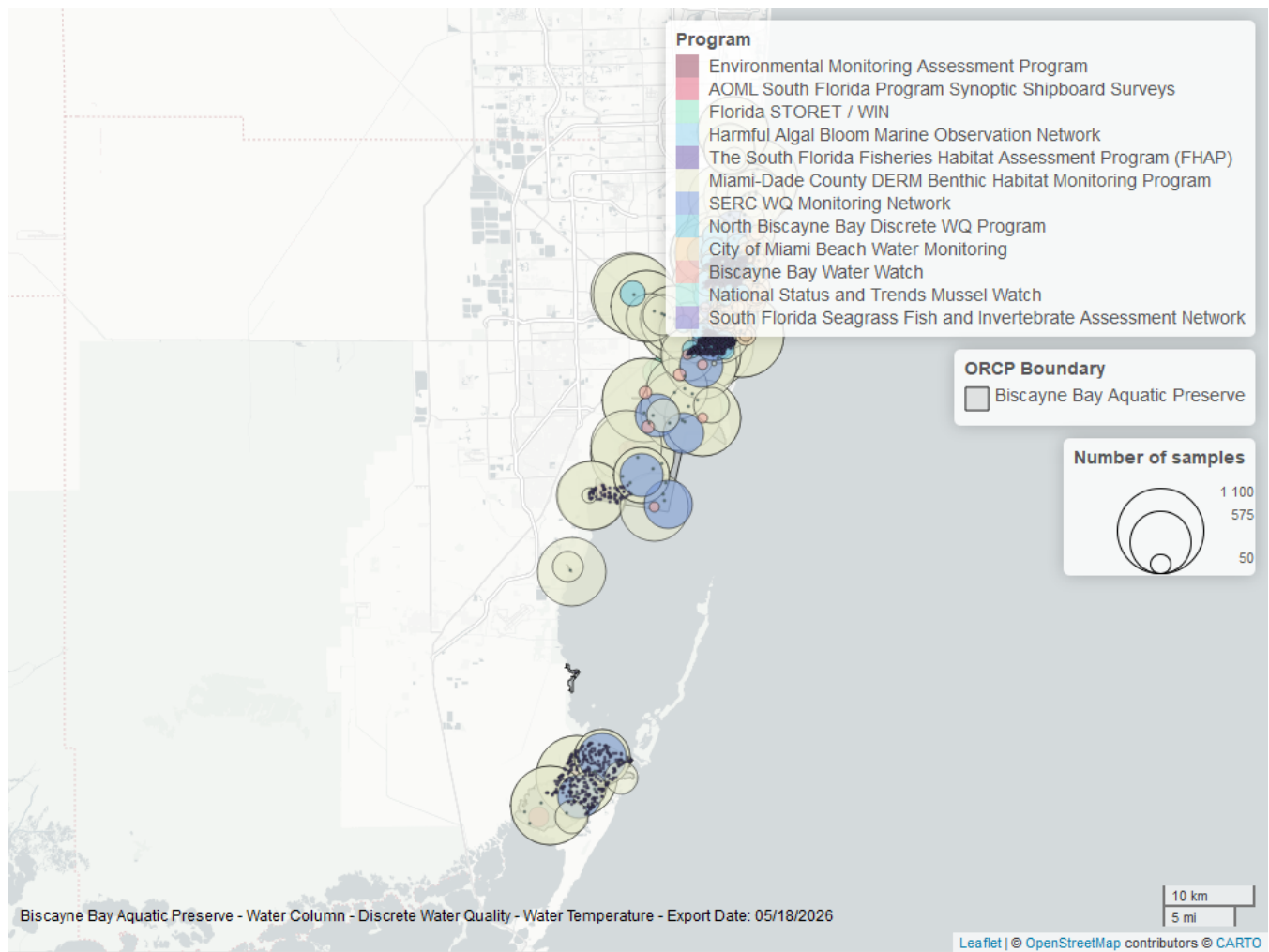


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 25: Programs contributing data for Water Temperature

ProgramID	N_Data	YearMin	YearMax
4018	38829	1994	2025
509	3316	1993	2008
4058	2694	2016	2025
965	2514	2005	2011
5026	1982	2019	2025
5002	1143	2001	2023
3	392	2002	2012
4049	271	2005	2008
102	267	1995	1996
4057	168	2015	2019
95	9	2012	2015
115	3	2004	2004

Program names:

4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³

509 - SERC Water Quality Monitoring Network⁴
4058 - City of Miami Beach Water Monitoring⁹
965 - South Florida Seagrass Fish and Invertebrate Assessment Network¹³
5026 - North Biscayne Bay Discrete Water Quality Program¹
5002 - Florida STORET / WIN²
3 - Atlantic Oceanographic and Meteorological Laboratory (AOML) South Florida Program Synoptic Shipboard Surveys¹⁴
4049 - The South Florida Fisheries Habitat Assessment Program (FHAP)¹⁰
102 - National Status and Trends Mussel Watch¹²
4057 - Biscayne Bay Water Watch¹¹
95 - Harmful Algal Bloom Marine Observation Network¹⁵
115 - Environmental Monitoring Assessment Program⁸

Water Quality - Continuous

The following files were used in the continuous analysis:

- *Combined_WQ_WC_NUT_cont_Chlorophyll_a_Uncorrected_for_Pheophytin-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen_Saturation-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Fluorescent_Dissolved_Organic_Matter-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_pH-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Salinity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Specific_Conductivity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Turbidity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Water_Temperature-2026-Mar-06.txt*

Continuous monitoring locations in Biscayne Bay Aquatic Preserve

Table 26: Station overview for Continuous parameters by Program

<i>ProgramID</i>	<i>ProgramLocationID</i>	<i>Years of Data</i>	<i>Use in Analysis</i>	<i>Parameters</i>
5077	BBBB14	7	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
5077	BBCWA4	4	FALSE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
5077	BBJT71	7	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
5077	BBLR03	6	TRUE	Dissolved Oxygen Saturation, Dissolved Oxygen
5077	BBLR03	7	TRUE	pH , Turbidity , Water Temperature , Salinity , Specific Conductivity
5077	BBMRDW	5	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
5077	BBMRRB	4	FALSE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
5077	BBMRSC	5	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity

Program names:

5077 - Biscayne Bay Aquatic Preserves Continuous Water Quality Monitoring¹⁶

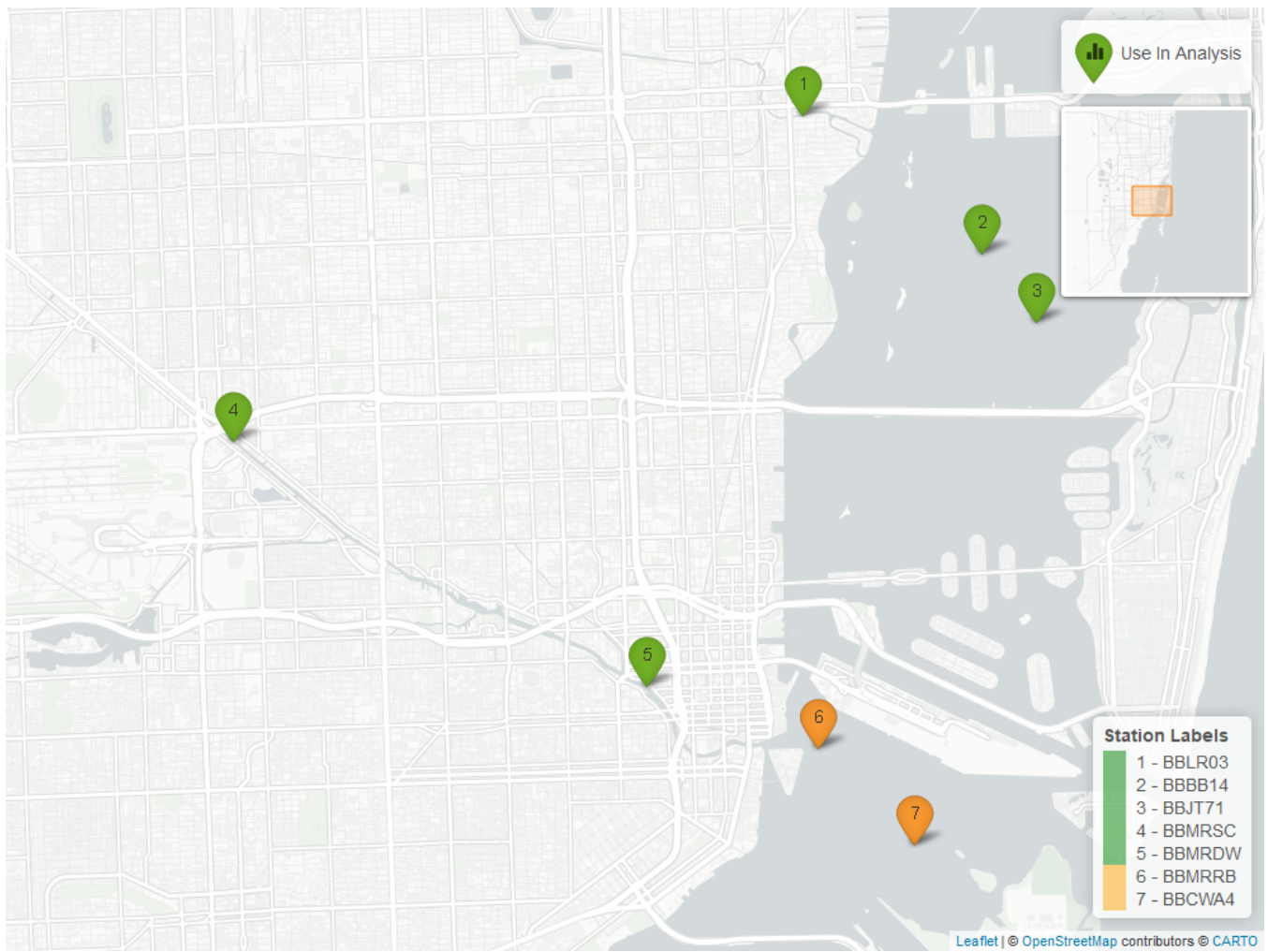


Figure 21: Map showing continuous water quality sampling locations within the boundaries of *Biscayne Bay Aquatic Preserve*. Sites marked as *Use In Analysis* (green) are featured in this report.

pH - Continuous

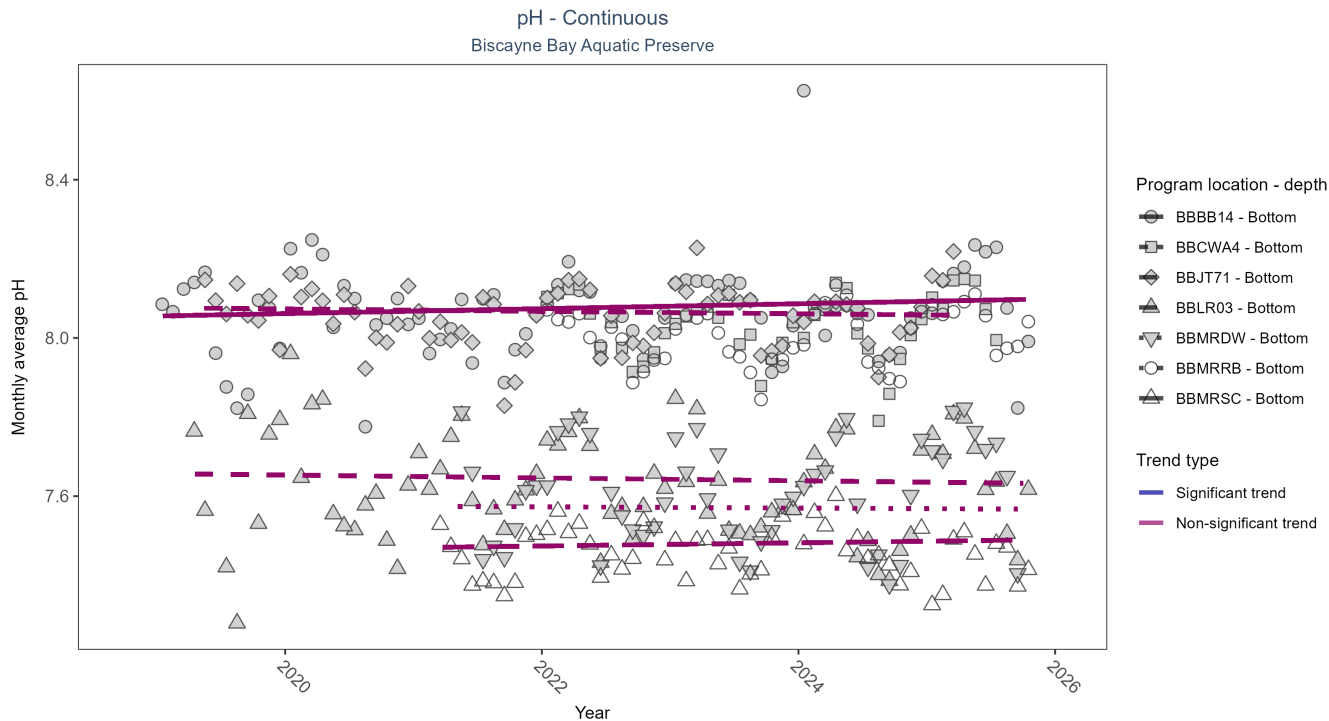


Figure 22: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 27: Seasonal Kendall-Tau Results for pH - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBBB14	No significant trend	205693	7	2019 - 2025	8.1	0.1	8.06	0.01	0.34
BBCWA4	Insufficient data to calculate trend	112102	4	2022 - 2025	8.0	-	-	-	-
BBJT71	No significant trend	195449	7	2019 - 2025	8.1	-0.07	8.08	0	0.51
BBLR03	No significant trend	195530	7	2019 - 2025	7.6	-0.05	7.66	0	0.67
BBMRDW	No significant trend	145249	5	2021 - 2025	7.6	-0.03	7.57	0	0.93
BBMRRB	Insufficient data to calculate trend	124839	4	2022 - 2025	8.0	-	-	-	-
BBMRSC	No significant trend	150527	5	2021 - 2025	7.4	0.03	7.47	0	0.7

No detectable change in monthly average pH was observed at five locations. There was insufficient data to fit a model for two locations.

Dissolved Oxygen Saturation - Continuous

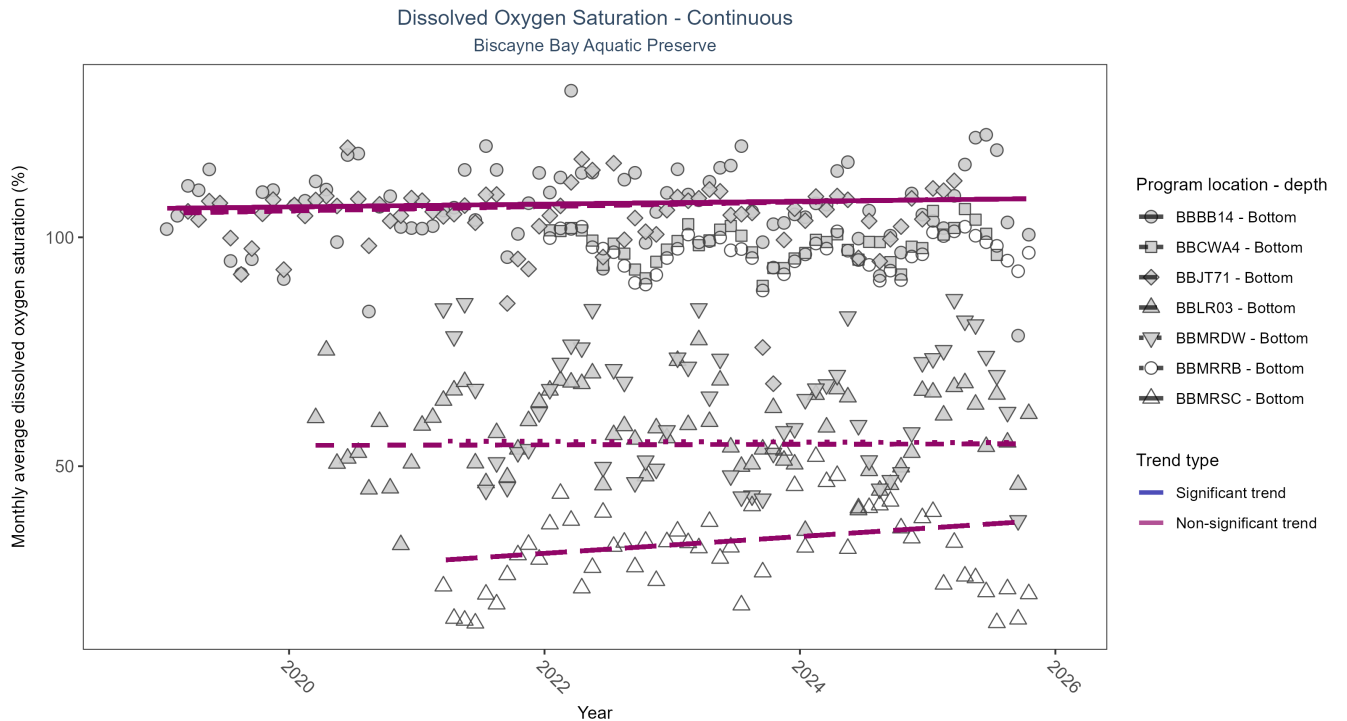


Figure 23: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 28: Seasonal Kendall-Tau Results for Dissolved Oxygen Saturation - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBBB14	No significant trend	209506	7	2019 - 2025	104.6	0.08	106.33	0.31	0.44
BBCWA4	Insufficient data to calculate trend	112145	4	2022 - 2025	97.8	-	-	-	-
BBJT71	No significant trend	208190	7	2019 - 2025	102.0	0.13	105.26	0.49	0.19
BBLR03	No significant trend	177332	6	2020 - 2025	57.7	0.03	54.54	0.06	0.86
BBMRDW	No significant trend	150198	5	2021 - 2025	62.5	-0.02	55.49	-0.07	0.94
BBMRRB	Insufficient data to calculate trend	124795	4	2022 - 2025	96.1	-	-	-	-
BBMRSC	No significant trend	151340	5	2021 - 2025	29.3	0.21	29.12	1.84	0.11

No detectable change in monthly average dissolved oxygen saturation was observed at five locations. There was insufficient data to fit a model for two locations.

Dissolved Oxygen - Continuous

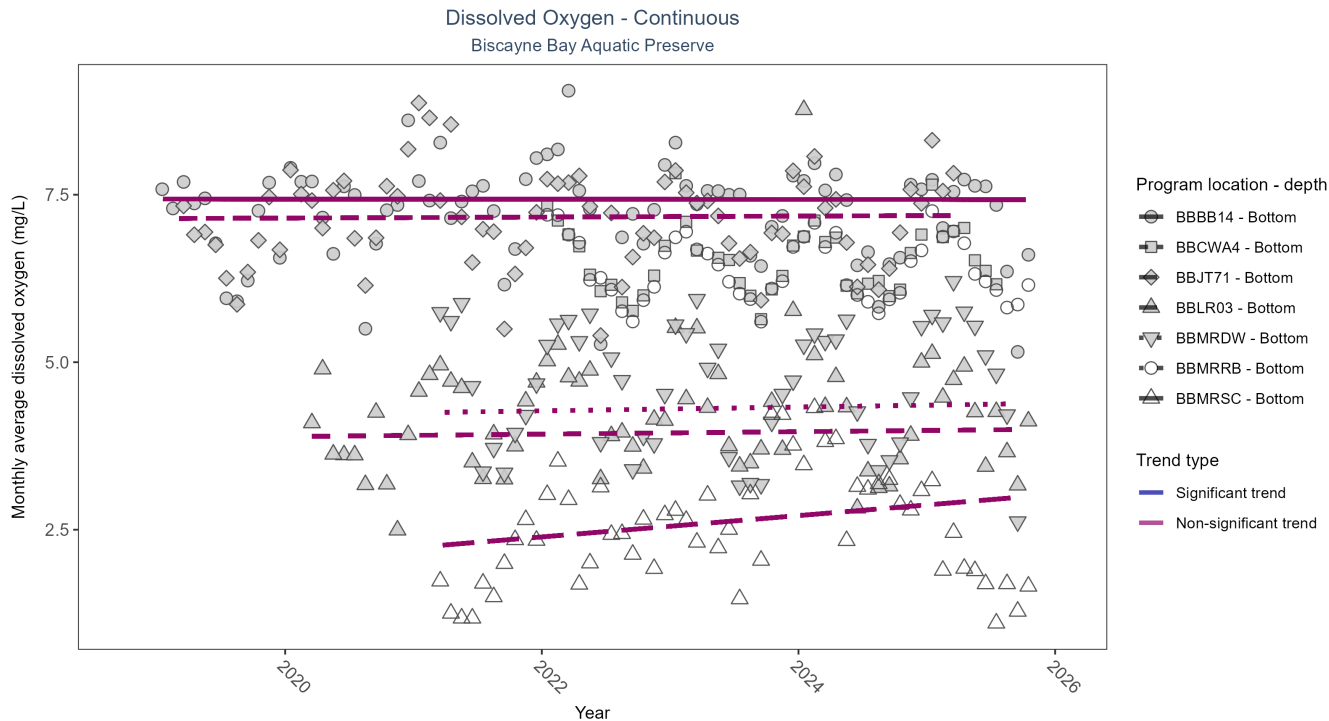


Figure 24: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 29: Seasonal Kendall-Tau Results for Dissolved Oxygen - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBBB14	No significant trend	209766	7	2019 - 2025	7.2	-0.01	7.43	0	1
BBCWA4	Insufficient data to calculate trend	108422	4	2022 - 2025	6.5	-	-	-	-
BBJT71	No significant trend	207039	7	2019 - 2025	7.0	0.06	7.14	0.01	0.56
BBLR03	No significant trend	177333	6	2020 - 2025	4.2	0.06	3.89	0.02	0.68
BBMRDW	No significant trend	150197	5	2021 - 2025	4.7	0.1	4.25	0.03	0.58
BBMRRB	Insufficient data to calculate trend	124787	4	2022 - 2025	6.4	-	-	-	-
BBMRSC	No significant trend	151445	5	2021 - 2025	2.2	0.23	2.23	0.16	0.08

No detectable change in monthly average dissolved oxygen was observed at five locations. There was insufficient data to fit a model for two locations.

Turbidity - Continuous

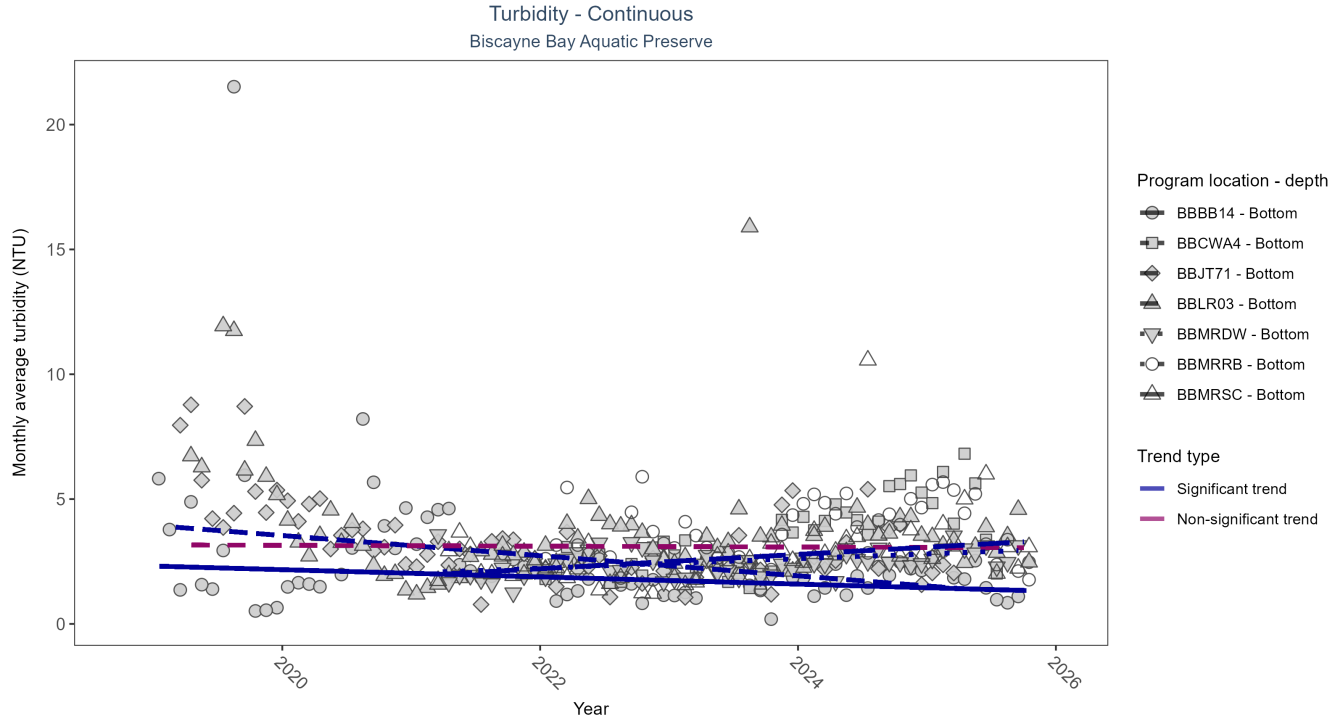


Figure 25: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 30: Seasonal Kendall-Tau Results for Turbidity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBB14	Significantly decreasing trend	195719	7	2019 - 2025	2	-0.22	2.32	-0.14	0.02
BBCWA4	Insufficient data to calculate trend	109259	4	2022 - 2025	3	-	-	-	-
BBJT71	Significantly decreasing trend	201128	7	2019 - 2025	3	-0.46	3.94	-0.4	0
BBLR03	No significant trend	202449	7	2019 - 2025	3	-0.01	3.16	-0.02	0.81
BBMRDW	Significantly increasing trend	147146	5	2021 - 2025	2	0.35	2.04	0.19	0.01
BBMRRB	Insufficient data to calculate trend	123415	4	2022 - 2025	3	-	-	-	-
BBMRSC	Significantly increasing trend	140544	5	2021 - 2025	2	0.41	1.92	0.29	0

At two program locations, monthly average turbidity increased by 0.19 NTU per year at one site and by 0.29 NTU per year at the other. At two program locations, monthly average turbidity decreased by 0.14 NTU per year at one site and by 0.4 NTU per year at the other. No detectable change in monthly average turbidity was observed at one location. There was insufficient data to fit a model for two locations.

Water Temperature - Continuous

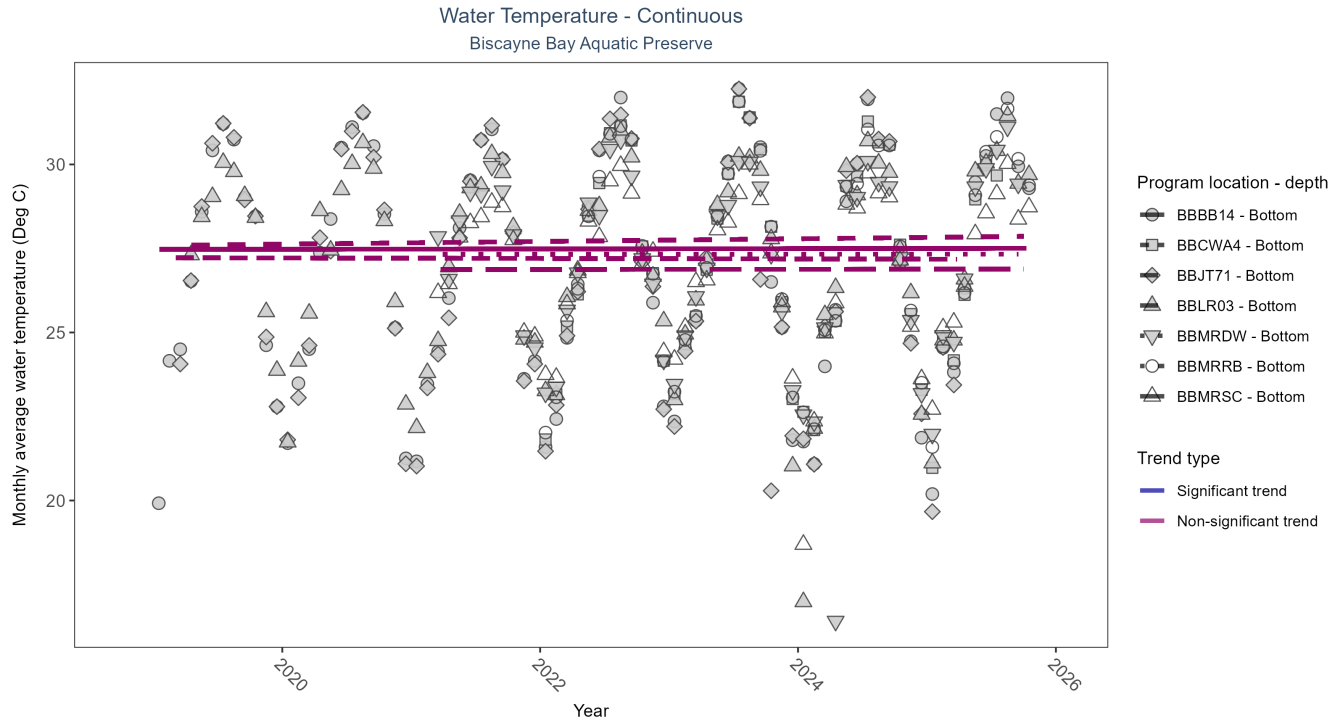


Figure 26: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 31: Seasonal Kendall-Tau Results for Water Temperature - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBBB14	No significant trend	209878	7	2019 - 2025	27.3	0.02	27.47	0	0.89
BBCWA4	Insufficient data to calculate trend	113090	4	2022 - 2025	27.1	-	-	-	-
BBJT71	No significant trend	208542	7	2019 - 2025	27.1	-0.03	27.22	-0.01	0.87
BBLR03	No significant trend	206286	7	2019 - 2025	27.9	0.04	27.59	0.04	0.67
BBMRDW	No significant trend	150209	5	2021 - 2025	27.6	-0.03	27.32	0	1
BBMRRB	Insufficient data to calculate trend	125603	4	2022 - 2025	27.3	-	-	-	-
BBMRSC	No significant trend	154183	5	2021 - 2025	27.3	-0.02	26.87	0	1

No detectable change in monthly average water temperature was observed at five locations. There was insufficient data to fit a model for two locations.

Salinity - Continuous

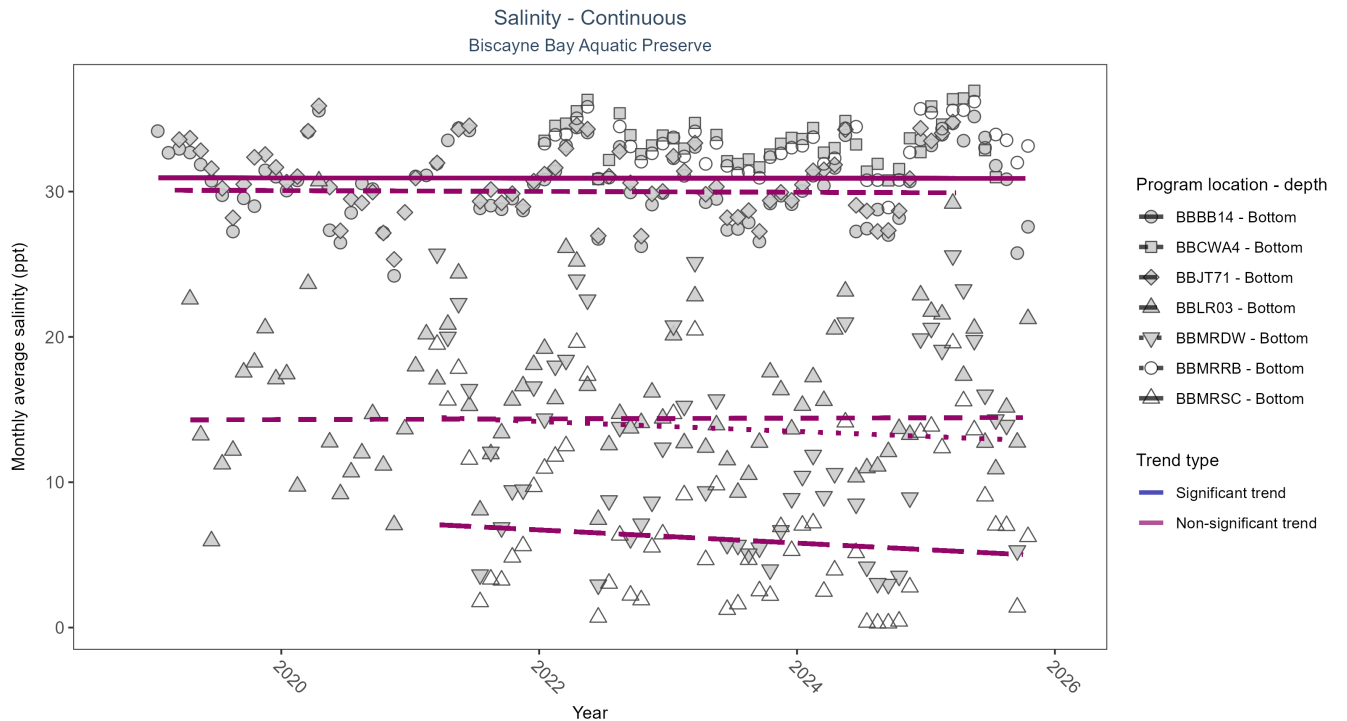


Figure 27: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 32: Seasonal Kendall-Tau Results for Salinity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBBB14	No significant trend	199380	7	2019 - 2025	30.6	-0.01	30.94	-0.01	1
BBCWA4	Insufficient data to calculate trend	106537	4	2022 - 2025	33.6	-	-	-	-
BBJT71	No significant trend	193275	7	2019 - 2025	31.0	-0.02	30.1	-0.03	0.87
BBLR03	No significant trend	197528	7	2019 - 2025	18.0	0.02	14.29	0.02	0.96
BBMRDW	No significant trend	146988	5	2021 - 2025	10.9	-0.17	14.52	-0.34	0.24
BBMRRB	Insufficient data to calculate trend	124795	4	2022 - 2025	33.3	-	-	-	-
BBMRSC	No significant trend	153400	5	2021 - 2025	5.1	-0.13	7.17	-0.45	0.32

No detectable change in monthly average salinity was observed at five locations. There was insufficient data to fit a model for two locations.

Specific Conductivity - Continuous

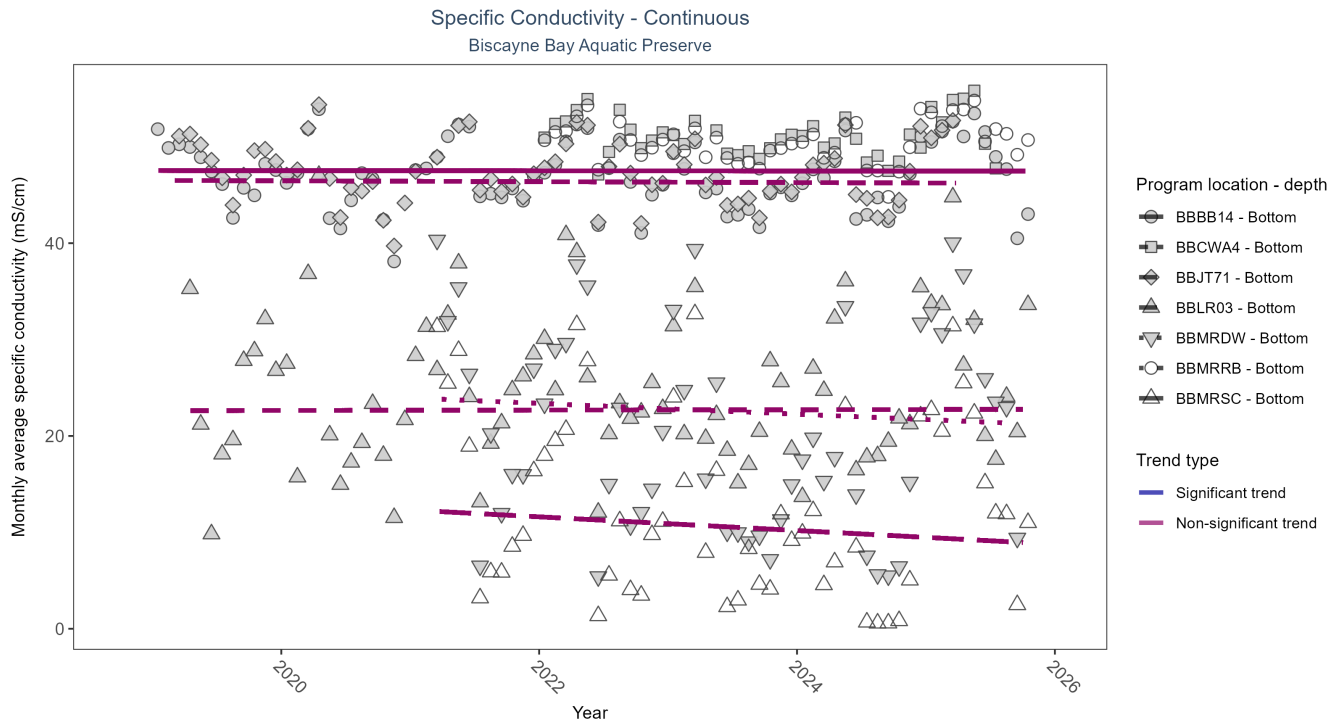


Figure 28: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 33: Seasonal Kendall-Tau Results for Specific Conductivity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBBB14	No significant trend	199381	7	2019 - 2025	47.11	-0.02	47.54	-0.01	0.85
BBCWA4	Insufficient data to calculate trend	106556	4	2022 - 2025	51.22	-	-	-	-
BBJT71	No significant trend	193275	7	2019 - 2025	47.63	-0.01	46.51	-0.05	0.96
BBLR03	No significant trend	199348	7	2019 - 2025	28.44	0.01	22.63	0.02	1
BBMRDW	No significant trend	146990	5	2021 - 2025	18.45	-0.17	23.94	-0.56	0.24
BBMRRB	Insufficient data to calculate trend	124795	4	2022 - 2025	50.80	-	-	-	-
BBMRSC	No significant trend	154166	5	2021 - 2025	8.98	-0.09	12.32	-0.71	0.49

No detectable change in monthly average specific conductivity was observed at five locations. There was insufficient data to fit a model for two locations.

Submerged Aquatic Vegetation

The data file used is: **All_SAV_Parameters-2026-Jun-04.txt**

Submerged aquatic vegetation (SAV) refers to plants and plant-like macroalgae species that live entirely underwater. The two primary categories of SAV inhabiting Florida estuaries are *benthic macroalgae* and *seagrasses*. They often grow together in dense beds or meadows that carpet the seafloor. *Macroalgae* include multicellular species of green, red and brown algae that often live attached to the substrate by a holdfast. They tend to grow quickly and can tolerate relatively high nutrient levels, making them a threat to seagrasses and other benthic habitats in areas with poor water quality. In contrast, *seagrasses* are grass-like, vascular, flowering plants that are attached to the seafloor by extensive root systems. *Seagrasses* occur throughout the coastal areas of Florida, including protected bays and lagoons as well as deeper offshore waters on the continental shelf. *Seagrasses* have taken advantage of the broad, shallow shelf and clear water to produce two of the most extensive seagrass beds anywhere in continental North America.

Parameters

Percent Cover measures the fraction of an area of seafloor that is covered by SAV, usually estimated by evaluating multiple small areas of seafloor. Percent cover is often estimated for total SAV, individual types of vegetation (seagrass, attached algae, drift algae) and individual species.

Frequency of Occurrence was calculated as the number of times a taxon was observed in a year divided by the number of sampling events, multiplied by 100. Analysis is conducted at the quadrat level and is inclusive of all quadrats (i.e., quadrats evaluated using Braun-Blanquet, modified Braun-Blanquet, and percent cover.”

Species

Turtle grass (*Thalassia testudinum*) is the largest of the Florida seagrasses, with longer, thicker blades and deeper root structures than any of the other seagrasses. It is considered a climax seagrass species.

Shoal grass (*Halodule wrightii*) is an early colonizer of vegetated areas and usually grows in water too shallow for other species except *widgeon grass*. It can often tolerate larger salinity ranges than other seagrass species. *Shoal grass* is characterized by thin, flat blades, that are narrower than *turtle grass* blades.

Manatee grass (*Syringodium filiforme*) is easily recognizable because its leaves are thin and cylindrical instead of the flat, ribbon-like form shared by many other seagrass species. The leaves can grow up to half a meter in length. *Manatee grass* is usually found in mixed seagrass beds or small, dense monospecific patches.

Widgeon grass (*Ruppia maritima*) grows in both fresh and salt water and is widely distributed throughout Florida’s estuaries in less saline areas, particularly in inlets along the east coast. This species resembles *shoal grass* in certain environments but can be identified by the pointed tips of its leaves.

Three species of *Halophila spp.* are found in Florida - **Star grass** (*Halophila engelmannii*), **Paddle grass** (*Halophila decipiens*), and **Johnson’s seagrass** (*Halophila johnsonii*). These are smaller, more fragile seagrasses than other Florida species and are considered ephemeral. They grow along a single long rhizome, with short blades. These species are not well-studied, although surveys are underway to define their ecological roles.

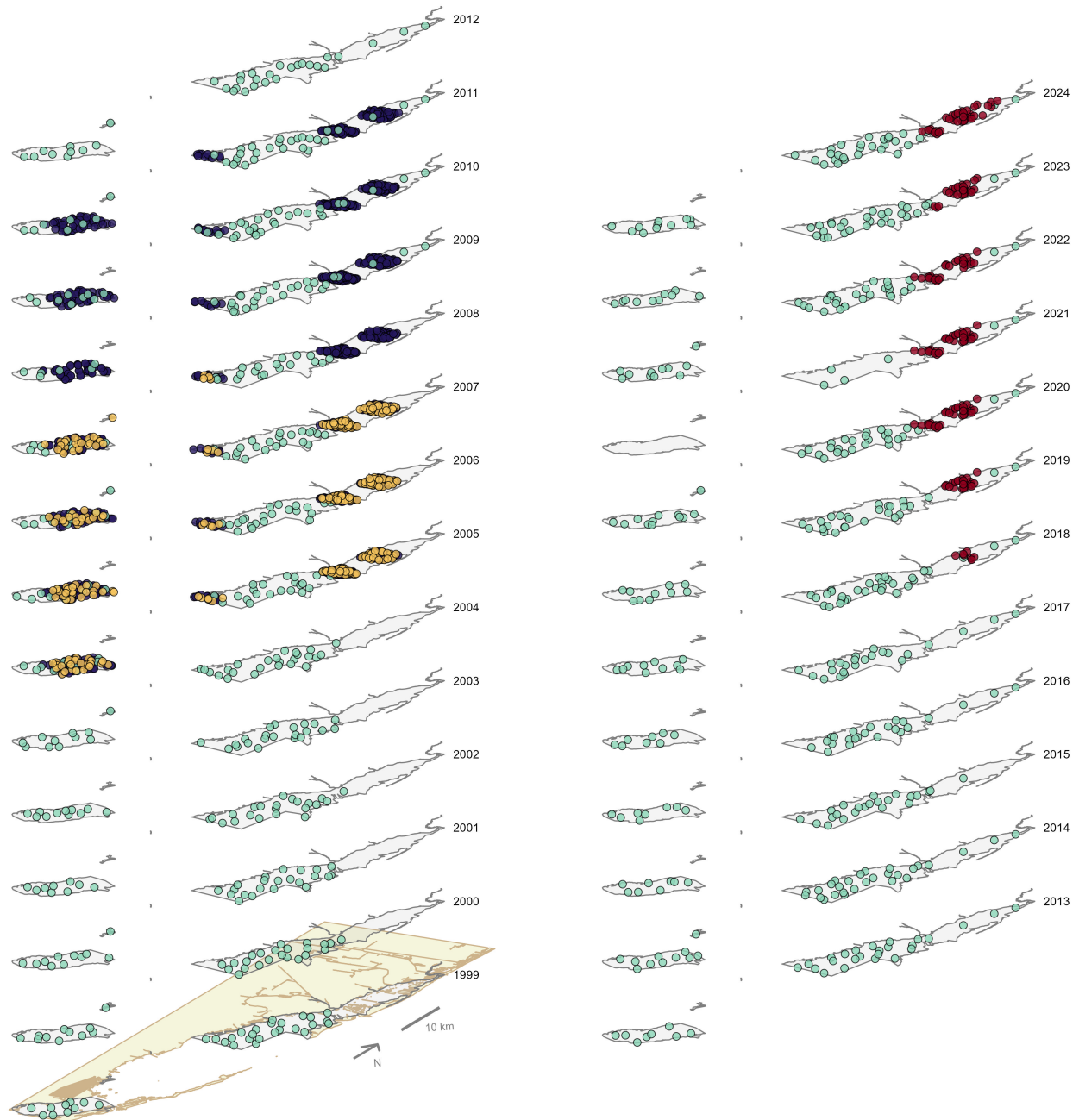
Notes

Star grass, *Paddle grass*, and *Johnson’s seagrass* will be grouped together and listed as **Halophila spp.** in the following managed areas. This is because several surveys did not specify to the species level:

- Banana River Aquatic Preserve
- Indian River-Malabar to Vero Beach Aquatic Preserve
- Indian River-Vero Beach to Ft. Pierce Aquatic Preserve
- Jensen Beach to Jupiter Inlet Aquatic Preserve
- Loxahatchee River-Lake Worth Creek Aquatic Preserve
- Mosquito Lagoon Aquatic Preserve

- Biscayne Bay Aquatic Preserve
- Florida Keys National Marine Sanctuary

Biscayne Bay Aquatic Preserve
SAV Percent Cover - Sample Locations



Program name

- Miami-Dade County DERM Benthic Habitat Monitoring Program
- South Florida Seagrass Fish and Invertebrate Assessment Network
- The South Florida Fisheries Habitat Assessment Program (FHAP)
- North Biscayne Bay Seagrass Loss Monitoring Program

Figure 29: Maps showing the temporal scope of SAV sampling sites within the boundaries of *Biscayne Bay Aquatic Preserve* by program name.

Click [here](#) to view spatio-temporal plots on GitHub.

Sampling locations by Program:

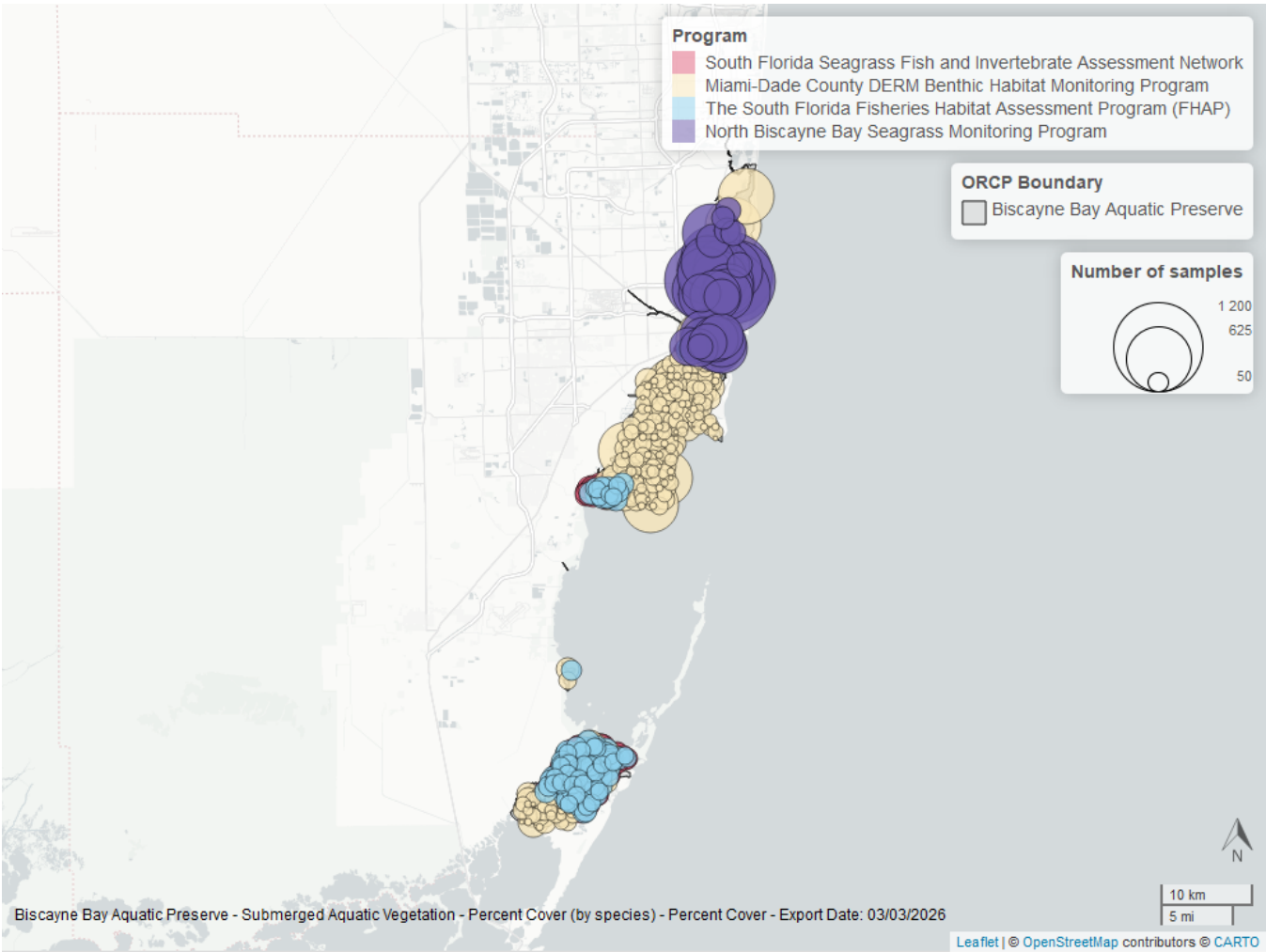


Figure 30: Map showing SAV sampling sites within the boundaries of *Biscayne Bay Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Table 34: Program Information for Submerged Aquatic Vegetation

<i>ProgramID</i>	<i>N-Data</i>	<i>YearMin</i>	<i>YearMax</i>	<i>method</i>	<i>Sample Locations</i>
965	71104	2005	2011	Braun Blanquet	94
4018	18876	1999	2024	Braun Blanquet	377
4049	18891	2005	2008	Braun Blanquet	273
5027	8250	2018	2024	Braun Blanquet	39
4018	966	1999	2007	Percent Cover	232
5027	8292	2018	2024	Percent Cover	39

Program names:

- 965 - South Florida Seagrass Fish and Invertebrate Assessment Network¹³
- 4018 - Miami-Dade County DERM Benthic Habitat Monitoring Program³
- 4049 - The South Florida Fisheries Habitat Assessment Program (FHAP)¹⁰
- 5027 - North Biscayne Bay Seagrass Monitoring Program¹⁷

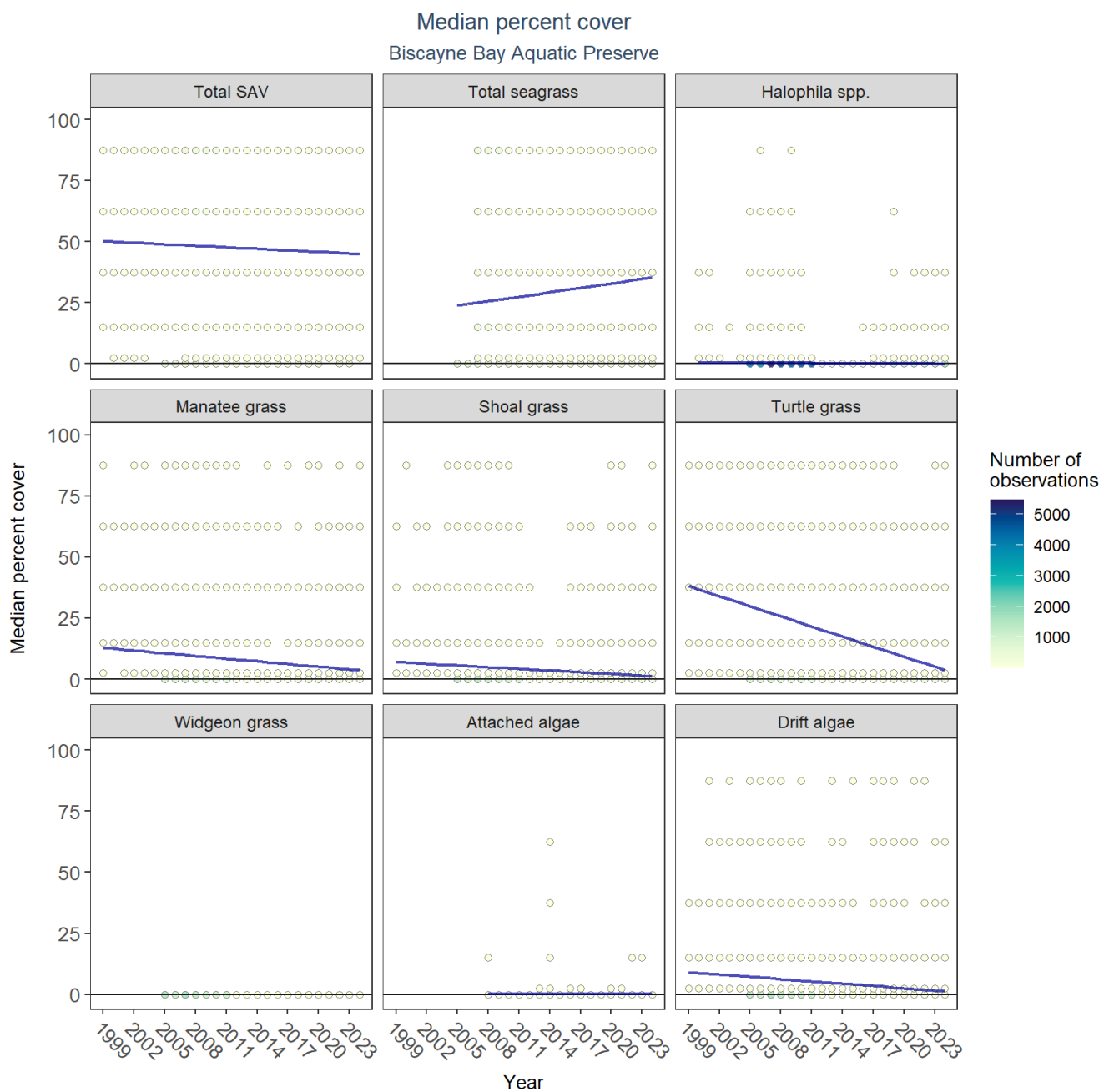


Figure 31: Scatter plots of median percent cover of submerged aquatic vegetation over time by group. Plots for time series that included five or more years of observations show the estimated trend as a blue line.

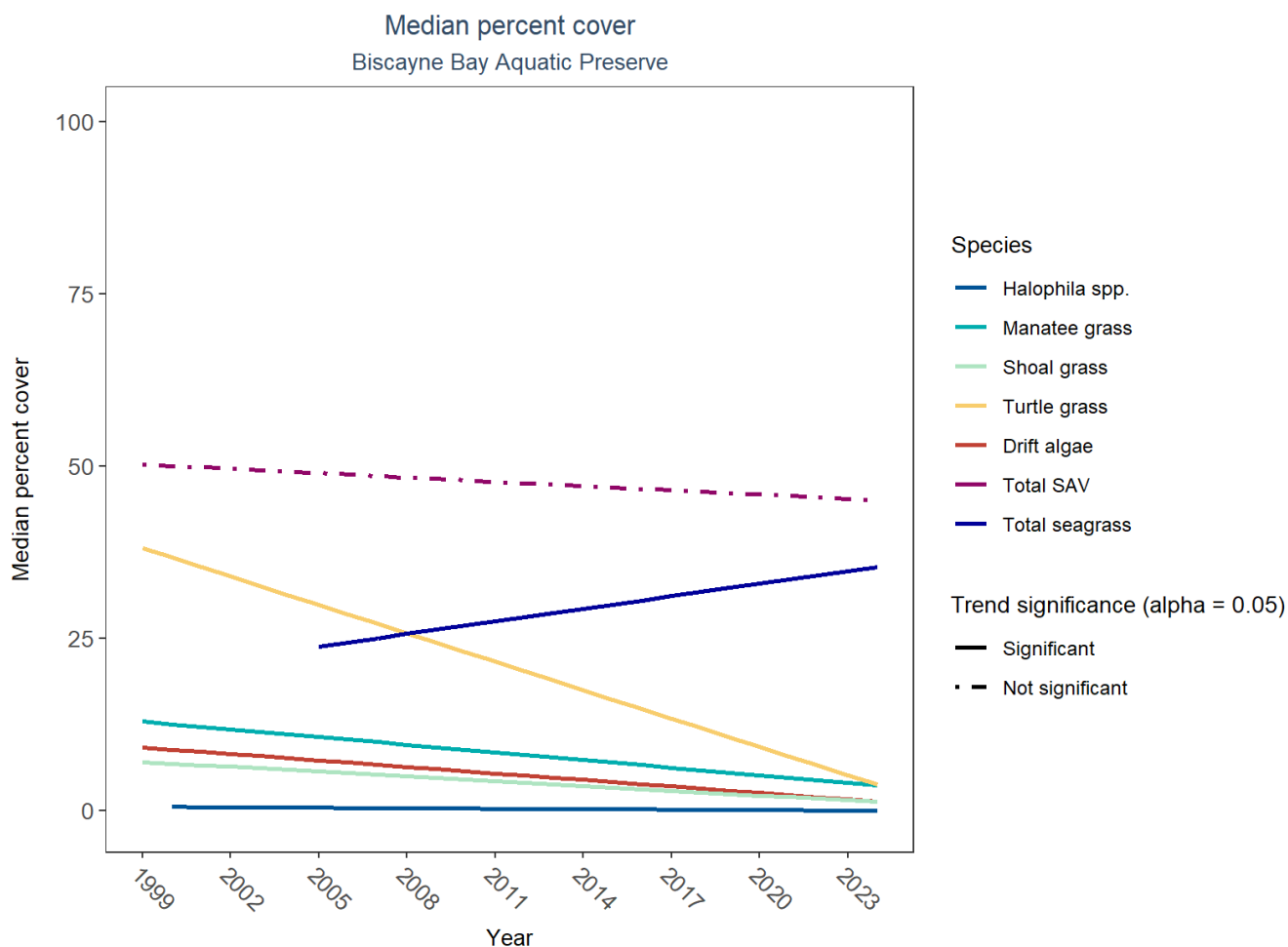


Figure 32: Trend lines of median percent cover of submerged aquatic vegetation over time for species that had five or more years of observations. Line type represents significance (solid) or non-significance (dashed) of the trends.

Table 35: Percent Cover Trend Analysis for Biscayne Bay Aquatic Preserve

<i>Species</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>LME Intercept</i>	<i>LME Slope</i>	<i>P</i>
Attached algae	No detectable trend	2008 - 2024	-0.0338584	0.0096341	0.1028814
Drift algae	Decreasing trend	1999 - 2024	10.7299717	-0.3136864	0.0000005
Shoal grass	Decreasing trend	1999 - 2024	8.2063045	-0.2315480	0.0000061
Halophila spp.	Decreasing trend	2000 - 2024	0.6686458	-0.0227177	0.0094048
Widgeon grass	Model did not fit the available data	2005 - 2024	-	-	-
Manatee grass	Decreasing trend	1999 - 2024	14.7500249	-0.3702606	0.0002814
Turtle grass	Decreasing trend	1999 - 2024	44.9650111	-1.3740768	0.0000000
Total SAV	No detectable trend	1999 - 2024	51.2374696	-0.2065003	0.2564943
Total seagrass	Increasing trend	2005 - 2024	17.1355555	0.6076970	0.0012280

An annual increase in percent cover was observed for total seagrass (0.61%). Annual decreases in percent cover were observed for Halophila spp. (-0.02%), manatee grass (-0.37%), shoal grass (-0.23%), turtle grass (-1.37%), and drift algae (-0.31%). No detectable change in percent cover was observed for total SAV and attached algae. The model could not be fitted for widgeon grass.

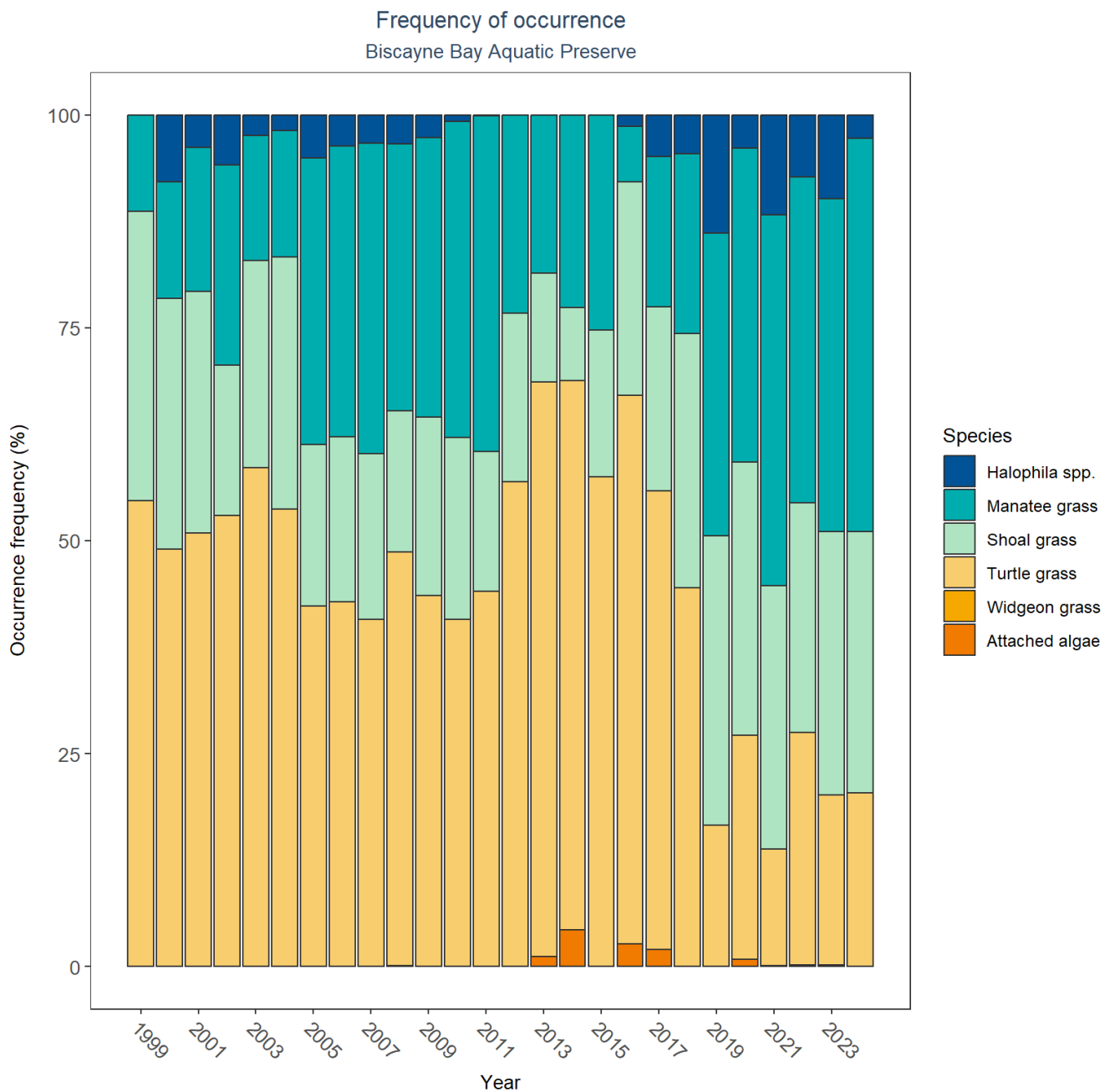


Figure 33: Bar plot of submerged aquatic vegetation species occurrence frequency over time by group.

SAV Water Column Analysis

The following parameters are available for Biscayne Bay Aquatic Preserve within the SAV_WC_Report:

- Colored Dissolved Organic Matter
- Chlorophyll a
- Dissolved Oxygen
- Dissolved Oxygen Saturation
- pH
- Salinity

- Secchi Depth
- Water Temperature
- Total Nitrogen
- Total Suspended Solids
- Turbidity

Access the reports here: [DRAFT_SAV_WC_Report.pdf](#)

Species list

Acanthophora muscoides ¹	Dictyosphaeria ¹	Porifera
Acanthophora sp. ¹	Dictyota spp. ¹	Pterocladia sanctarum ¹
Acetabularia sp. ¹	Drift red algae ¹	Rhipocephalus ¹
Acetabularia spp. ¹	Ernodesmis sp. ¹	Rhipocephalus phoenix ¹
Actiniaria	Galaxaura spp. ¹	Rhipocephalus spp. ¹
Agardhiella ramosissima ¹	Gracilaria sp. ¹	Ruppia maritima ¹
Amphiroa spp. ¹	Gracilaria spp. ¹	Sargassum sp. ¹
Anadyomene linkiana ¹	Grateloupia ¹	Sargassum spp. ¹
Anadyomene spp. ¹	Griffithsia ¹	Schizothrix calcicola
Anadyomene stellata ¹	Halimeda discoidea ¹	Scleractinia
Avrainvillea levis ¹	Halimeda incrassata ¹	Spermothamnion ¹
Batophora oerstedii ¹	Halimeda opuntia ¹	Spyridia filamentosa ¹
Batophora spp. ¹	Halimeda spp. ¹	Syringodium filiforme ¹
Brown algae ¹	Halodule wrightii ¹	Thalassia testudinum ¹
Bryopsis ¹	Halophila decipiens ¹	Total brown algae ¹
Calcareous green algae ¹	Halophila engelmannii ¹	Total calcareous green algae ¹
Caulerpa cupressoides ¹	Halophila johnsonii ¹	Total green algae ¹
Caulerpa sertularioides ¹	Halophila sp. ¹	Total macroalgae ¹
Caulerpa spp. ¹	Hydrozoa	Total other green algae ¹
Centroceras sp. ¹	Hypnea ¹	Total other red algae ¹
Ceramium ¹	Hypoglossum ¹	Total SAV ¹
Chaetomorpha linum ¹	Jania spp. ¹	Total seagrass ¹
Champia parvula ¹	Laurencia spp. ¹	Turbinaria turbinata
Chara spp. ¹	Neomeris ¹	Udotea ¹
Chondria capillaris ¹	Octocorallia	Udotea spp. ¹
Chondria spp. ¹	Other coral	Ulva sp. ¹
Cladocephalus ¹	Other green algae ¹	Unidentified mangrove
Cladophora ¹	Other red algae ¹	Unidentified species
Crustose coralline algae ¹	Padina gymnospora ¹	Unknown blue sponge
Dasya sp. ¹	Penicillus sp. ¹	Unknown orange encrusting sponge
Dasycladus ¹	Penicillus spp. ¹	Valonia ¹
Derbesia ¹	Polysiphonia sp. ¹	Wrightiella ¹

1 - Submerged Aquatic Vegetation

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